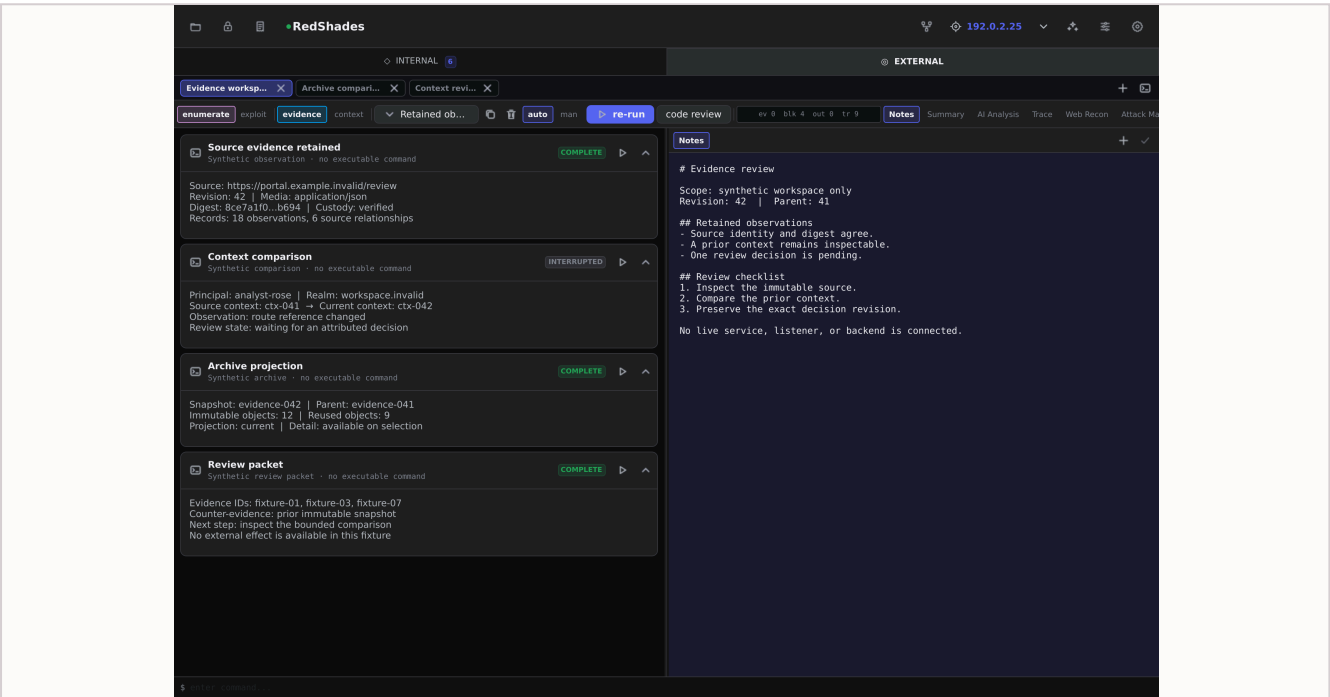


THE PRODUCT AND ENGINEERING PROPOSITION

RedShades. Intelligence with an operating system.

An 88-stage adaptive chain, custom capability generation, mid-run software repair and a native Qt workspace connect in one independently developed platform. RedShades combines deterministic orchestration with focused AI reasoning to carry complex security investigations across tools, branches and long-running sessions.



Actual Qt capture · 5 September 2026 · fictional, disconnected fixture. Presentation evidence only.

The central proposition is architectural: spend model capacity on interpretation and synthesis while software owns workflow, context, validation and execution policy. The 88 automatic declarations form a conditional graph; separate C2 stage families extend session work.

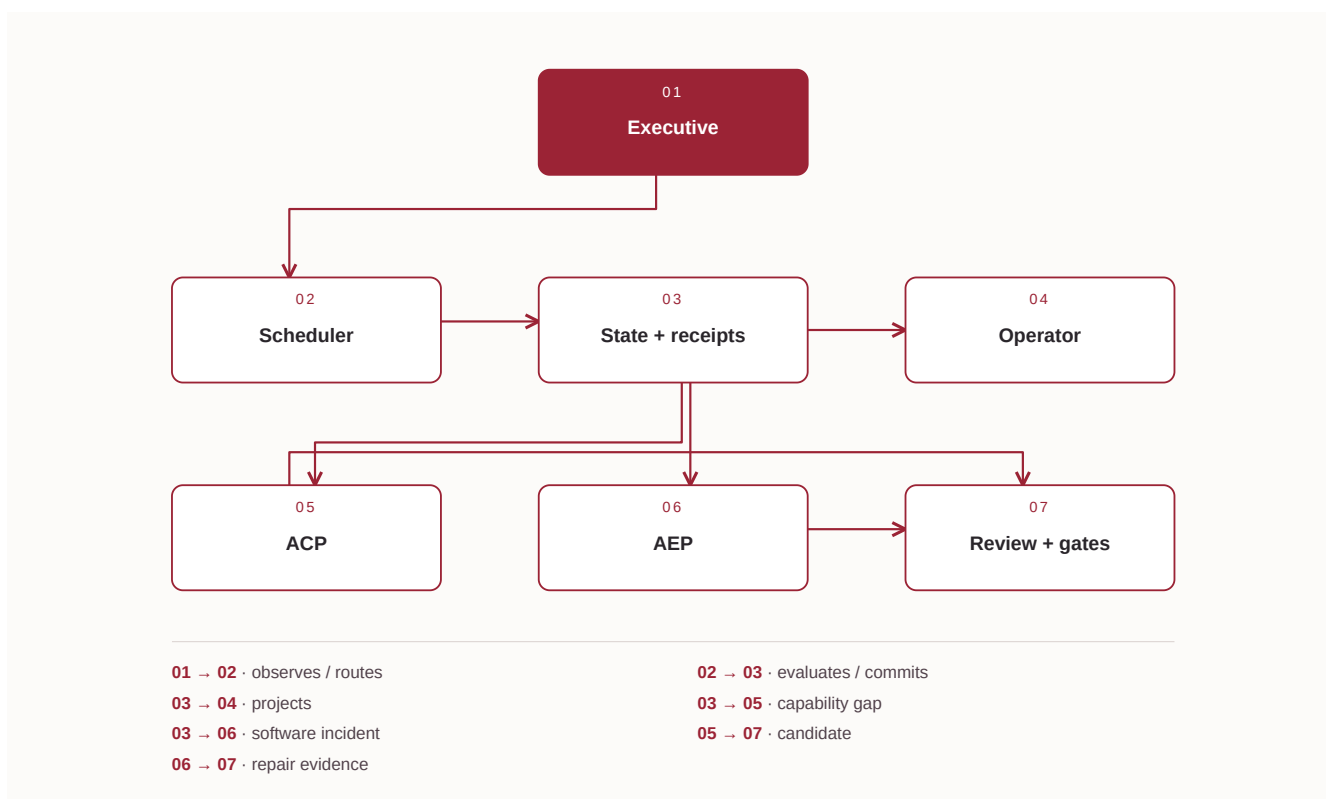
Implemented contract

Reference 01
authoritative engagement architecture

WHO DECIDES WHAT HAPPENS NEXT?

A chain that schedules, branches and adapts

The scheduler evaluates 88 automatic stage declarations against current evidence. Independent branches advance when their inputs are ready; loops revisit useful frontiers as observations change. The Executive maintains the wider view while ACP supplies missing capabilities and AEP repairs the underlying software.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Each role has a distinct responsibility: scheduling, supervision, bounded interpretation, capability generation or platform repair. Their shared records let the investigation progress beyond one prompt or one tool.

Implemented contract

Reference 02
subsystem glossary

PARENT CONTEXT, CHILD SCOPE, CONNECTED PROGRESS

Pivot into a child investigation

A newly available route can open a child subchain with its own scope and execution context. Parent and child retain their relationship, so a deeper investigation remains part of the same attributed engagement.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

Frontiers track eligible and waiting work. Branch-local decisions allow unrelated work to continue; parent completion accounts for attached investigations. This is a graph of evolving work rather than a fixed checklist.

Implemented contract

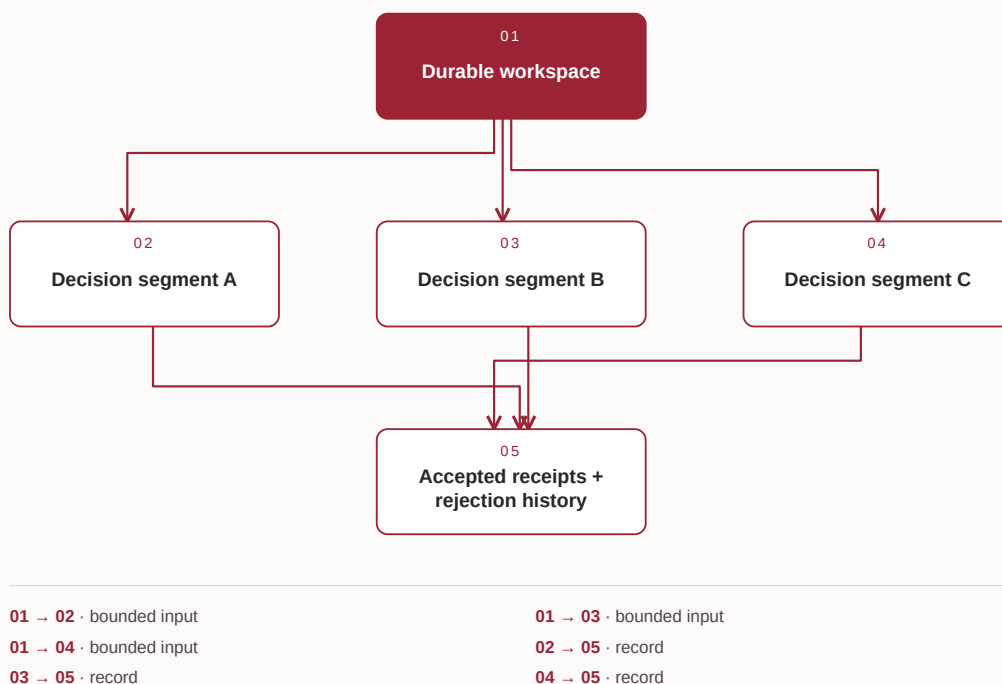
Reference 03

authoritative engagement architecture

HOW AI ALLOCATION CONNECTS TO PRODUCT VALUE

70% deterministic. 30% neural.

RedShades uses this ratio as a design philosophy: dependable software surrounds the places where model judgment adds value. Scheduling, state, policy and validation provide structure; models interpret ambiguity, develop candidates and guide complex engineering work.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

The ratio is a design framing, not a measured code, runtime or success percentage. The intended economic advantage is selective intelligence: compact specialists for bounded decisions, persistent sessions for work that benefits from continuity.

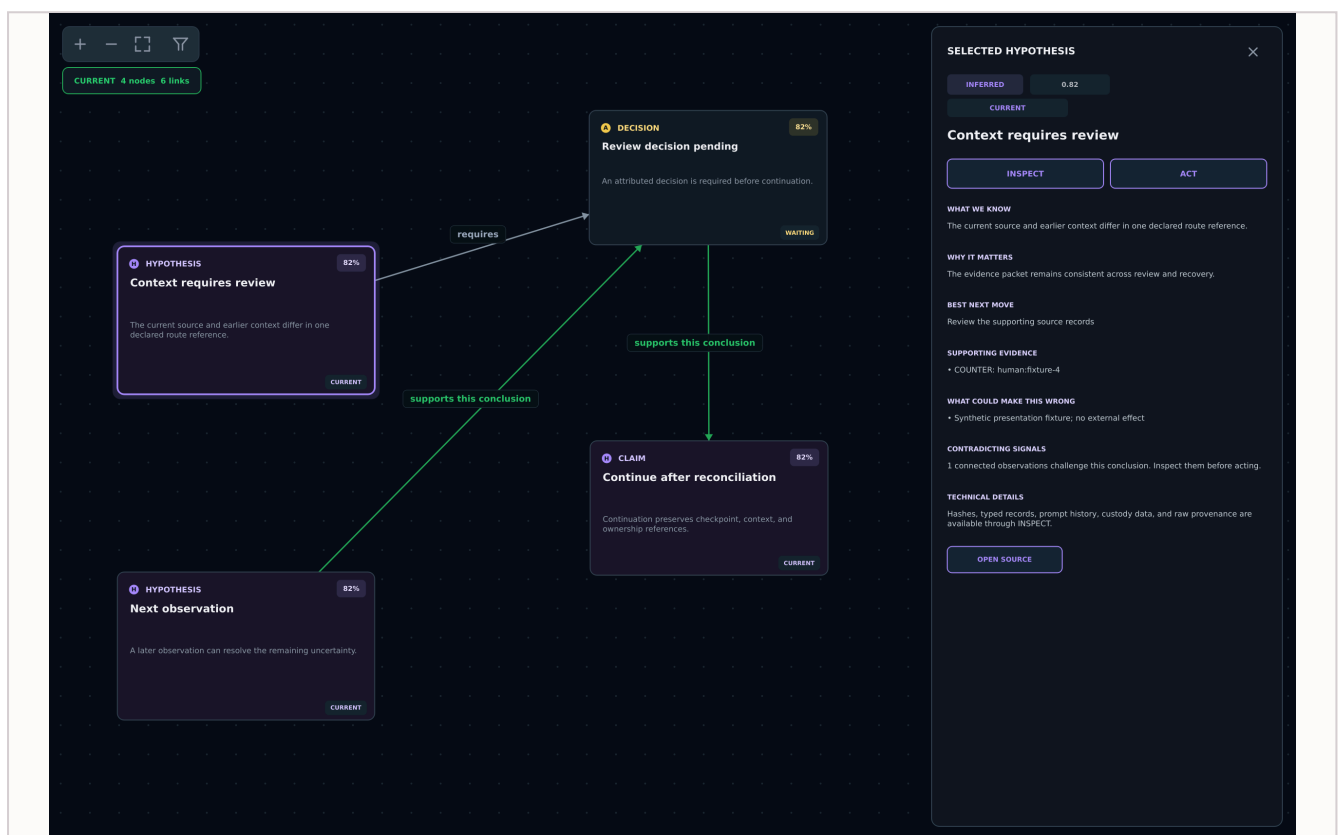
Design philosophy

Reference 04
acp vertical session

WHAT BELONGS TOGETHER?

An engagement memory shared by every subsystem

Evidence, identities, routes, artifacts, capabilities and outcomes are linked in versioned engagement records. The scheduler, model workspaces and operator views can work from the same account of the investigation.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

Revision-bound records preserve the inputs behind each decision. Canonical state supplies the architecture for coordinated work across tools and sessions; source references document the implementation boundaries.

Implemented contract

Reference 05

authoritative engagement architecture

WHAT MAKES EVIDENCE REUSABLE?

From tool output to reusable evidence

Integrated tools contribute observations with their source, context and interpretation retained. Source references make an accepted result useful to the next stage, the next model segment and the operator inspecting the decision.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

Evidence categories support correlation without flattening distinct candidates into one summary. This is the connection between broad tool coverage and coherent automated reasoning.

Implemented contract

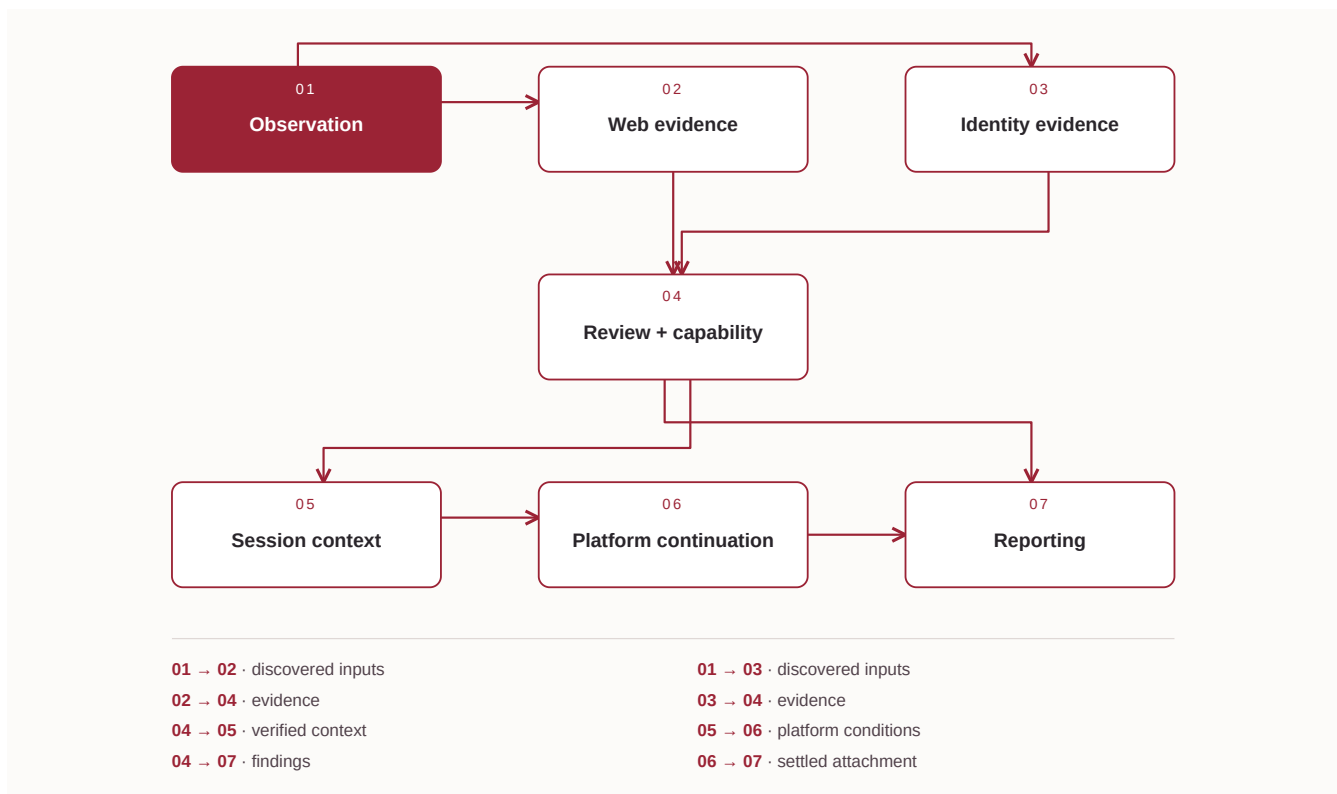
Reference 06

canonical specialist contracts

WHAT DOES THE STAGE COUNT DESCRIBE?

88 automatic stages, one adaptive graph

Discovery, service and identity analysis, web investigation, artifact analysis, session work and reporting are represented in 88 automatic stage declarations. A separate set of 13 C2 declarations adds common and platform-specific session families.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Prerequisites, evidence, platform selection and operator policy select the actual path. The stage atlas provides every automatic declaration with its source label and input/output role; 88 does not describe a mandatory linear execution sequence.

Implemented contract

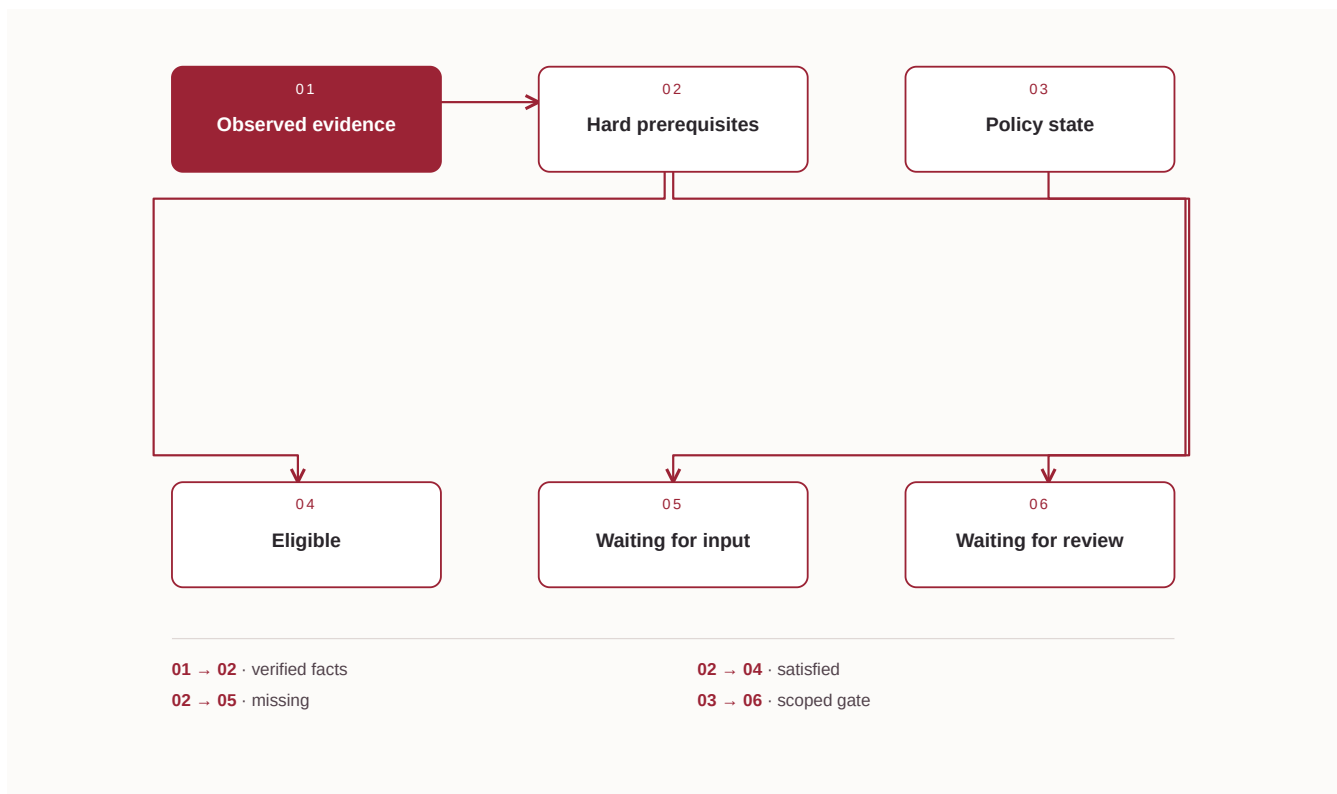
Reference 07

`semantic_registry.go + chain_runtime.go`

HOW THE GRAPH AVOIDS INVENTING AUTHORITY

A scheduler that understands relationships

The Execution Graph distinguishes scheduling dependencies from evidence correlations. Phase barriers, declared branch dependencies and scoped eligibility control work, while context and evidence links enrich the operator's view.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

The scheduler evaluates a bounded stage view. Dynamic endpoint and gate nodes remain useful to the operator without multiplying the work required to decide which stage is ready.

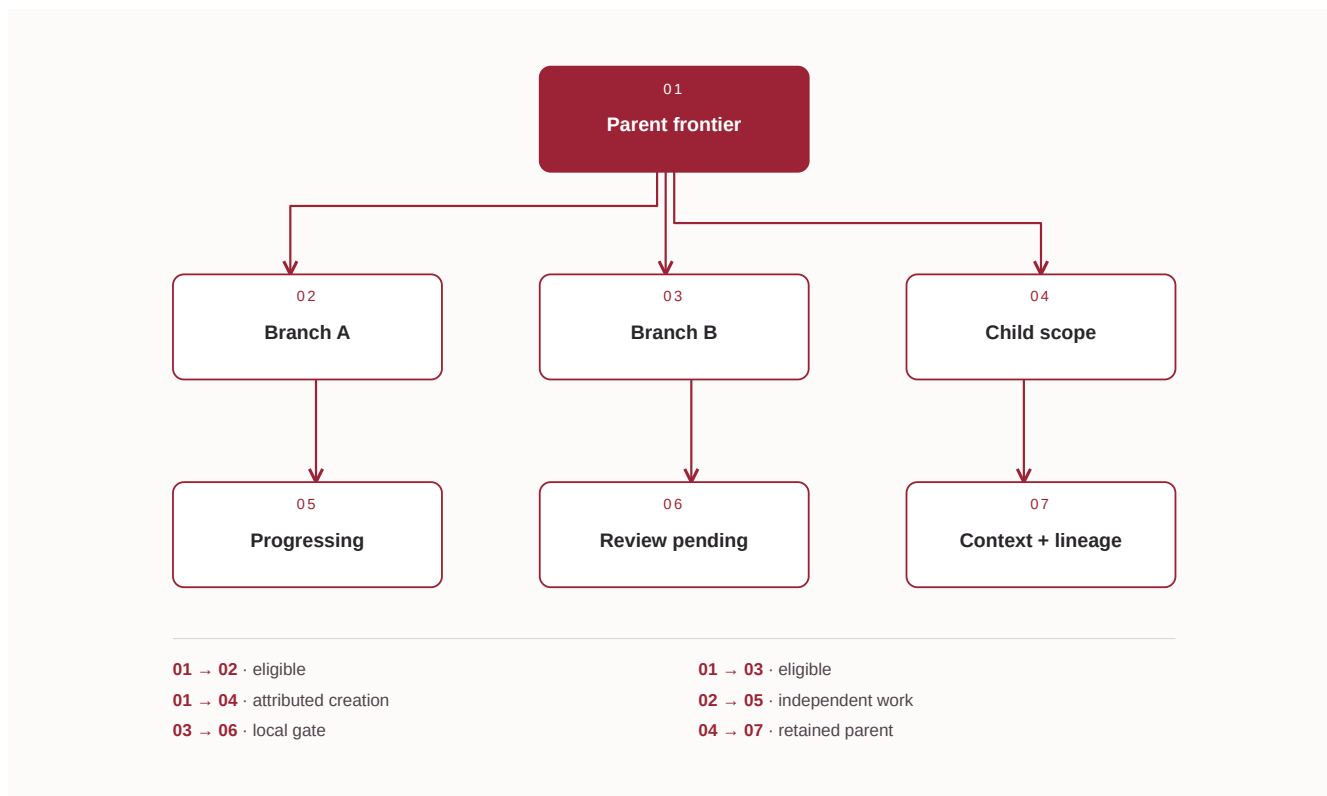
Implemented contract

Reference 08
execution graph lifecycle

HOW BRANCHES PROGRESS INDEPENDENTLY

Keep independent branches moving

The scheduler separates branch-local gates from a global pause. A web review, identity question or pending prerequisite can wait in its own scope while other eligible work advances.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Loops and dormant branches respond to new evidence. The frontier records which input can change eligibility, giving long investigations a clear continuation point.

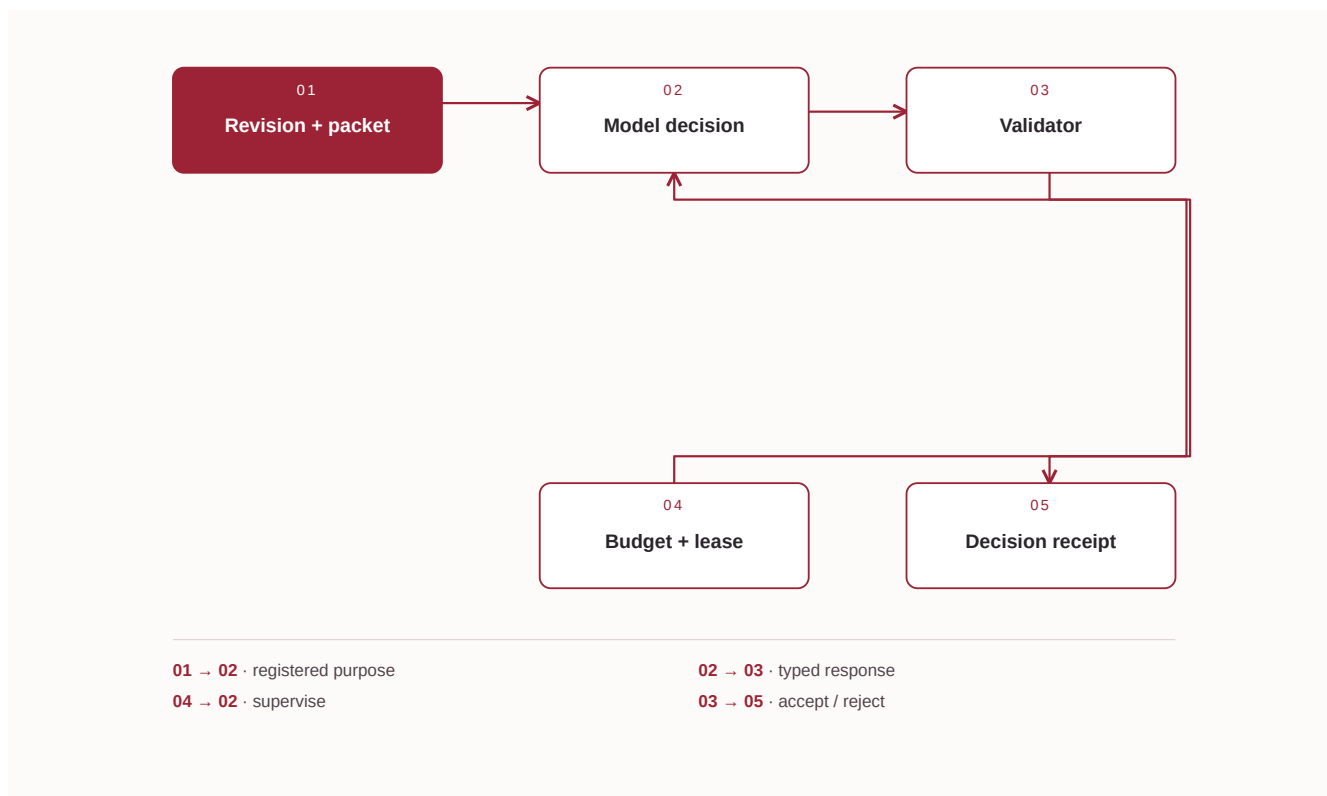
Implemented contract

Reference 09
execution graph lifecycle

WHAT SURROUNDS A DECISION?

Small calls with complete jobs

Bounded stateless AI calls receive a defined purpose and a structured evidence packet. Deterministic code checks the response, attributes the model and retains the accepted result. This is the intended home for compact, fine-tuned specialist models.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Token accounting, capacity controls and response validation surround the call. A 1B-class specialist is a deployment objective for suitable bounded tasks; the catalog does not claim a trained model bank or measured equivalence to a larger model.

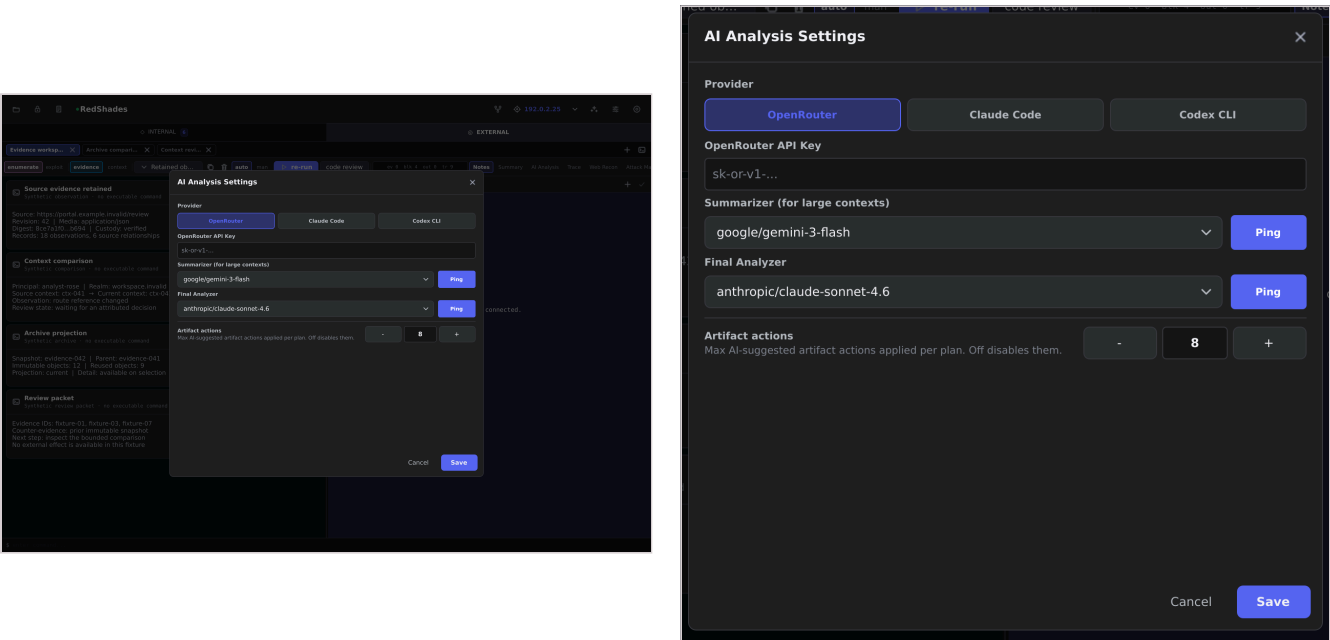
Implemented contract

Reference 10
authoritative engagement architecture

WHAT IS THE EXECUTIVE WATCHING?

An Executive that retains the wider mission

The Executive observes progress across stages, prioritizes attention and routes capability gaps or engineering incidents. Its persistent supervisory context complements the short, focused calls used inside individual boundaries.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Overview and detail crop from identical source pixels. Presentation evidence only.

Executive, ACP and AEP workspaces preserve artifacts and checkpoints for longer tasks. Model capacity can be allocated by responsibility, while the deterministic controller retains authority over the workflow.

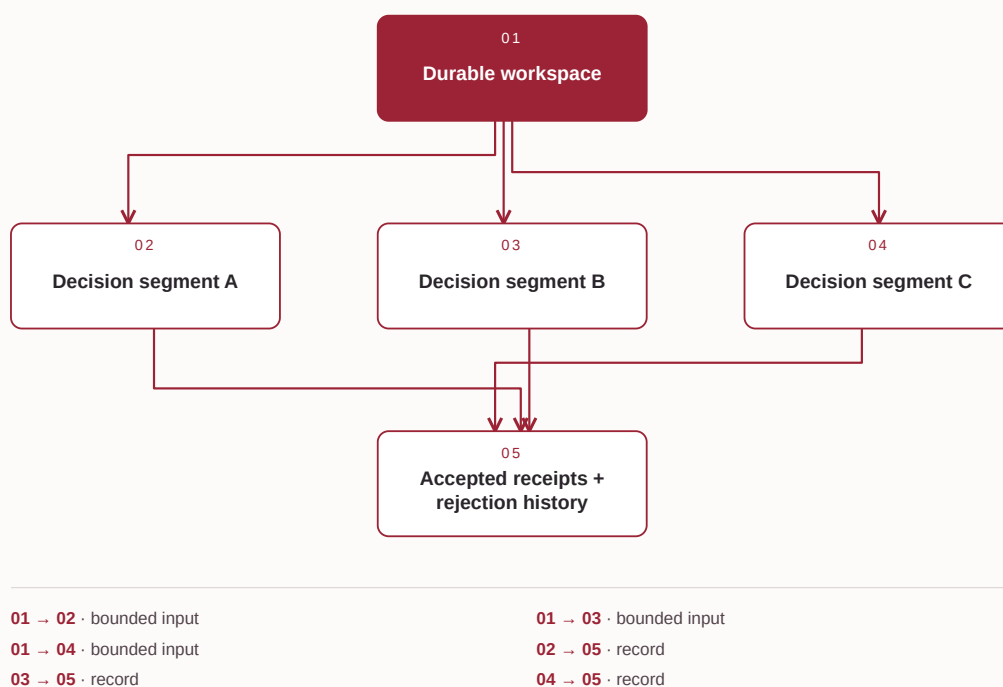
Implemented contract

Reference 11
executive teacher babysitter plan

HOW COULD A COMPACT MODEL CONTRIBUTE?

Compact models, durable engineering context

Persistent Executive, ACP and AEP sessions can use more capable compact models for synthesis and repair. Custom context management carries accepted artifacts, prior review and progress records between segments, so the task does not depend on an indefinitely growing conversation.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Model selection is configurable. Terra-low is a documented operating policy in the project history; alternative model deployments require their own evaluation. The architecture targets strong outcomes from efficient models without asserting an unmeasured quality or speed equivalence.

Implemented model boundaries

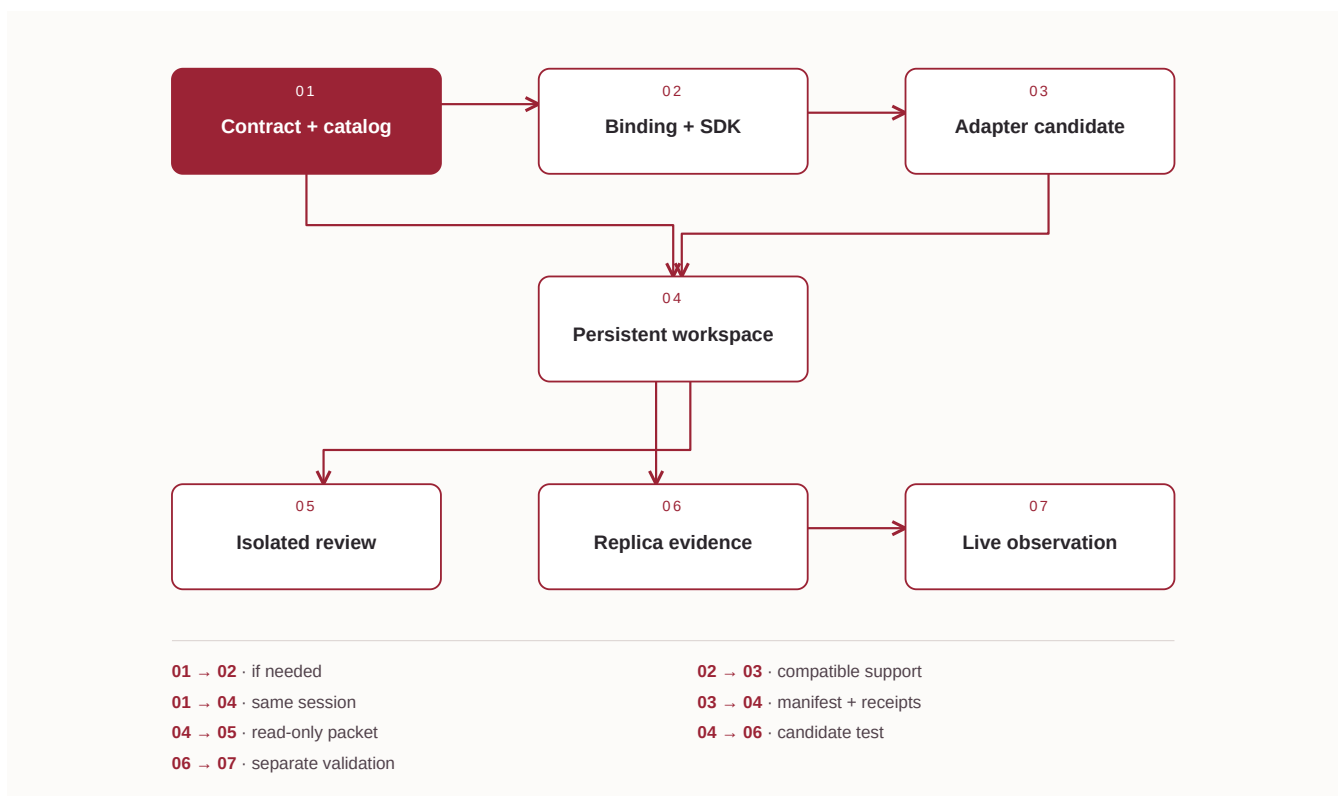
Reference 12

canonical specialist contracts

WHY PERSIST A GENERATION SESSION?

ACP builds evidence-bound capabilities

The Adaptive Capability Pipeline develops custom capability candidates from observed evidence and source mechanics. A persistent workspace owns binding, SDK selection or extension, adapter generation, review and replica validation.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Compatible catalog artifacts and SDKs can skip generation segments. New work retains its contract and review history, connecting flexibility with reusable, reviewed artifacts.

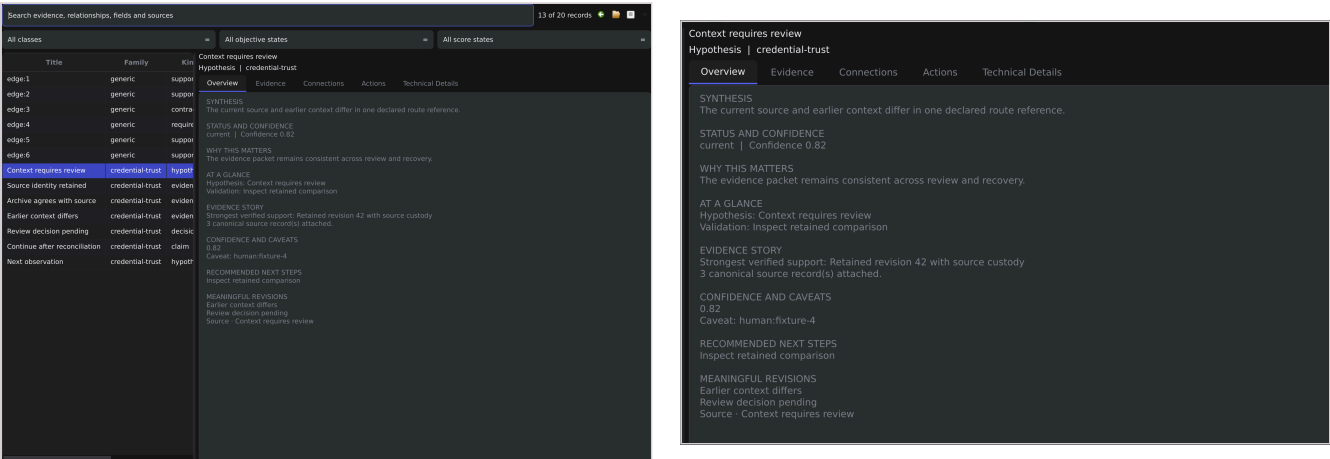
Implemented contract

Reference 13
 acp vertical session

WHAT CROSSES THE GENERATION BOUNDARY?

From an advisory to a reviewed candidate

ACP connects source findings and advisories with typed capability contracts, SDK extensions and reviewed adapters. Where a ready-made proof of concept is absent, the workbench can retain a candidate for controlled evaluation.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Overview and detail crop from identical source pixels. Presentation evidence only.

Candidate source, metadata and tests are published as a digest-bound set.
Admission and isolated review establish the boundary between generated material and an approved artifact.

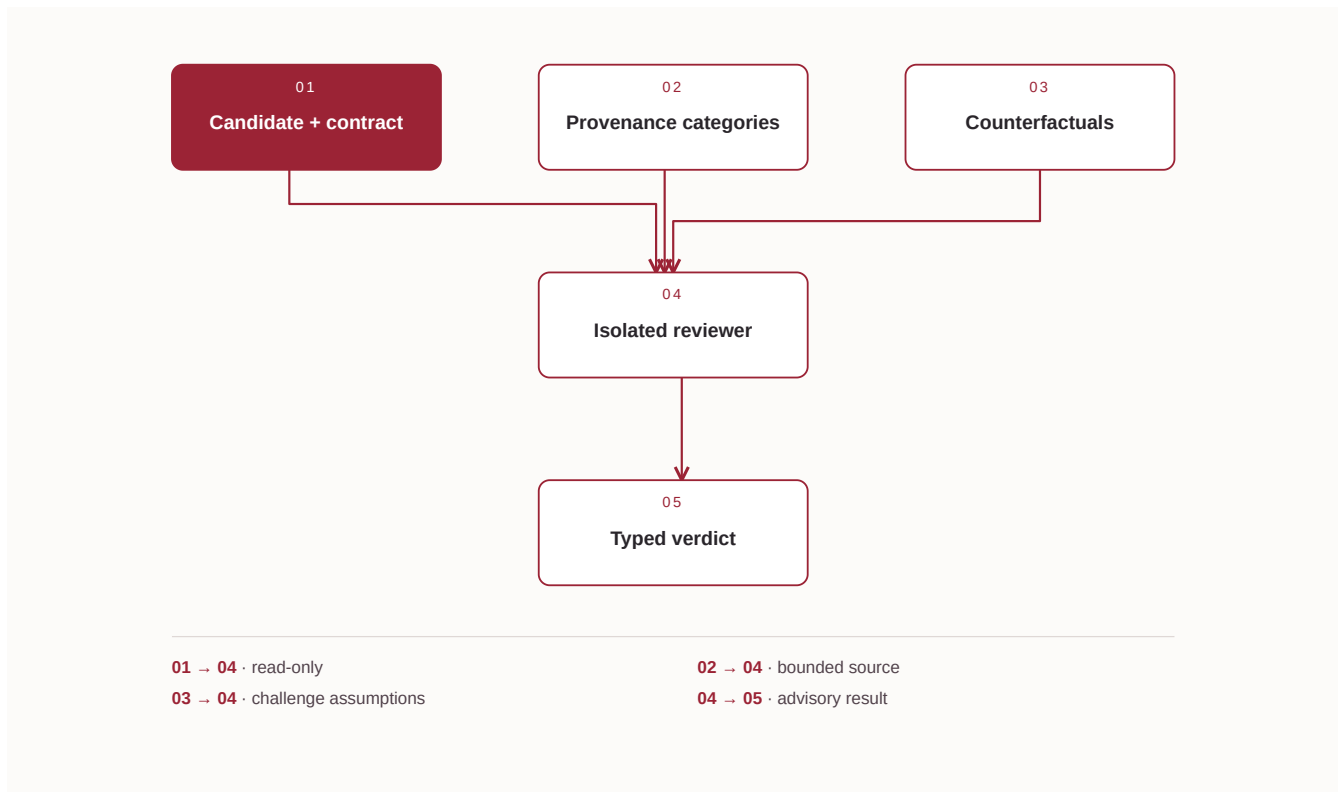
Implemented contract

Reference 14
acp vertical session

WHY ISOLATE THE CONTAMINATION REVIEWER?

Review is built into capability production

An isolated reviewer evaluates the candidate against its contract and provenance. Counterfactuals test whether changed names, ordering and evidence still support the same capability, helping turn a one-off result into a reusable component.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

The reviewer uses a read-only packet and returns a typed verdict to admission. Generation, review and execution authority remain separately owned.

Implemented contract

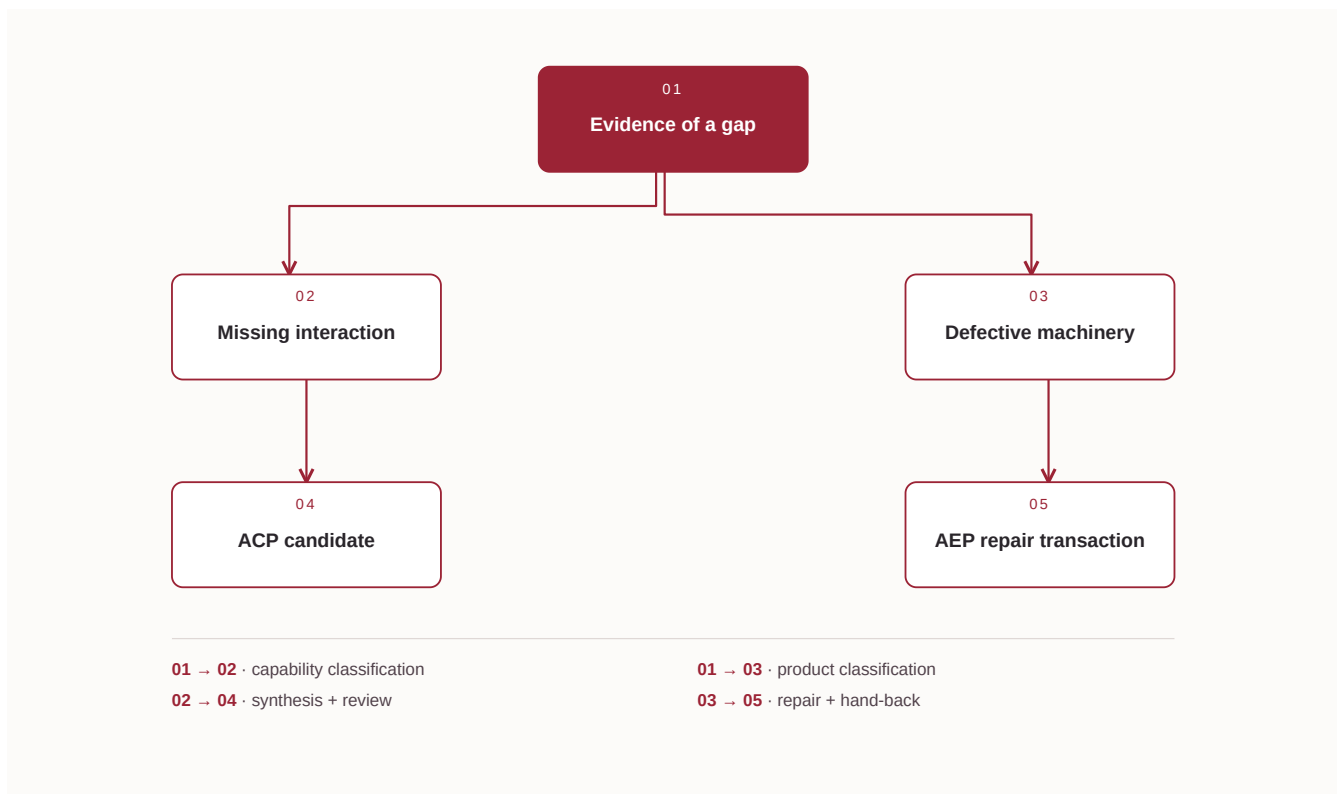
Reference 15

acp vertical session

WHERE ACP ENDS AND AEP BEGINS

AEP repairs software while the run is in flight

The Autonomic Engineering Plane owns diagnosis, a persistent repair session, scoped source changes, tests, build admission, hot replacement and verified hand-back. It addresses defects in the platform machinery while ACP addresses missing target interactions.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

The full repair transaction preserves the run's context through a provider replacement. Its acceptance is defined by validated hand-back, connecting software engineering work to continued execution.

Implemented contract

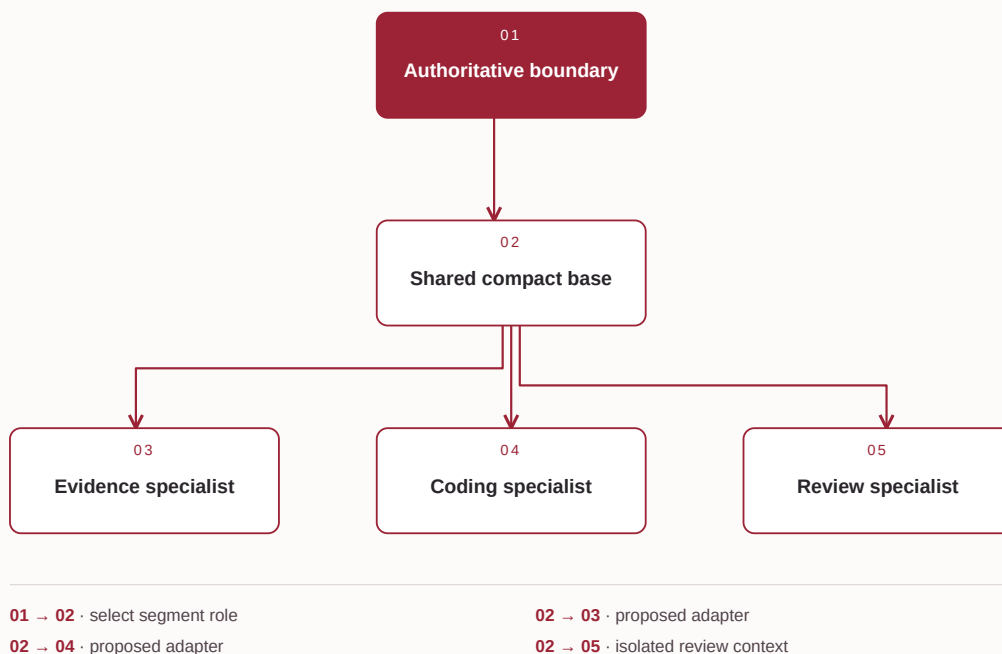
Reference 16

executive teacher babysitter plan

WHAT IS PROPOSED FOR MODEL ALLOCATION?

A proposed bank of 100 specialist adapters

The model direction is a compact shared base with a proposed bank of roughly 100 specialist QLoRA adapters, selected at deterministic task boundaries. Specialized short calls would share the surrounding evidence, workflow and validation architecture.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

This is proposed model architecture, not a claim that the bank is trained or deployed. Persistent synthesis sessions remain a complementary path for harder work; serving and cross-adapter context behavior require evaluation.

Proposed model architecture

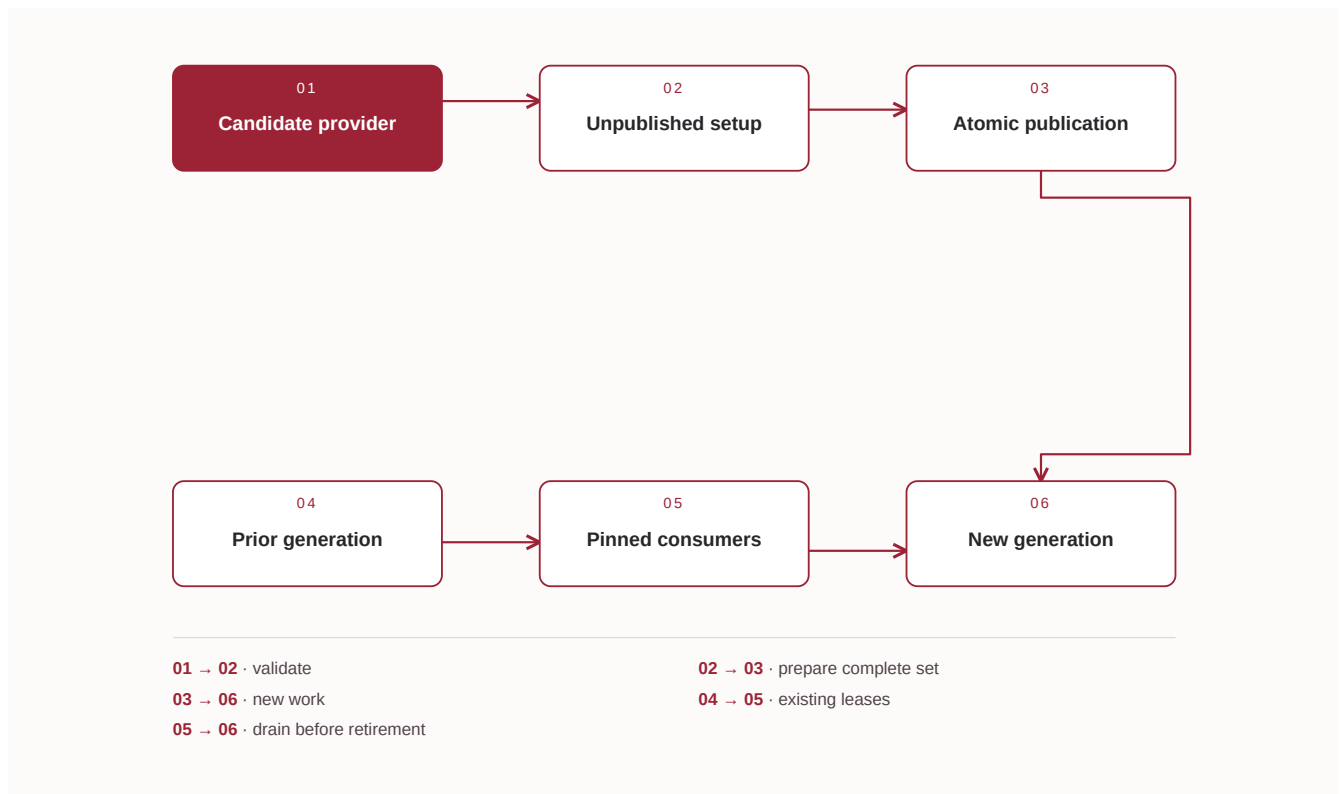
Reference 17

acp vertical session + design history

WHAT DOES THE COMPOSITION KERNEL OWN?

Hot replacement with continuity

The Runtime Composition Kernel publishes versioned providers and controls leases, rollback and cleanup. New work can adopt a replacement while existing work retains its implementation and engagement records.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

This is the runtime foundation beneath AEP hand-back. State remains outside replaceable behavior, giving repair a defined boundary rather than requiring a whole-workspace restart.

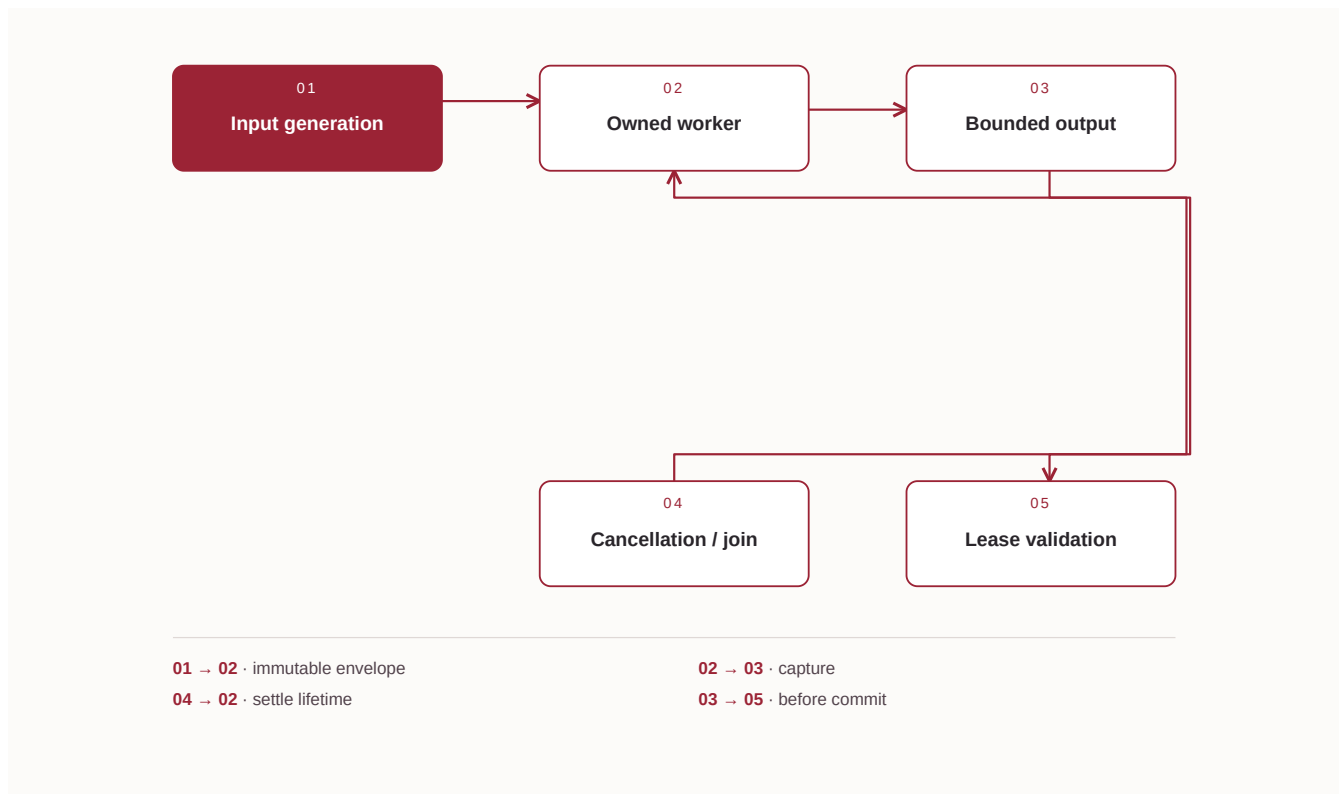
Implemented contract

Reference 18
runtime composition kernel

HOW IS STALE WORK REJECTED?

Workers return results under a lease

Isolated stage workers receive immutable execution envelopes. Results are accepted only against the current artifact, input and generation lease. That prevents an old worker from publishing a plausible result after its authority has expired.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

In-process handlers and isolated workers have different replacement boundaries. The catalog describes the common principle without implying that every component can be replaced at any arbitrary instruction.

Implemented contract

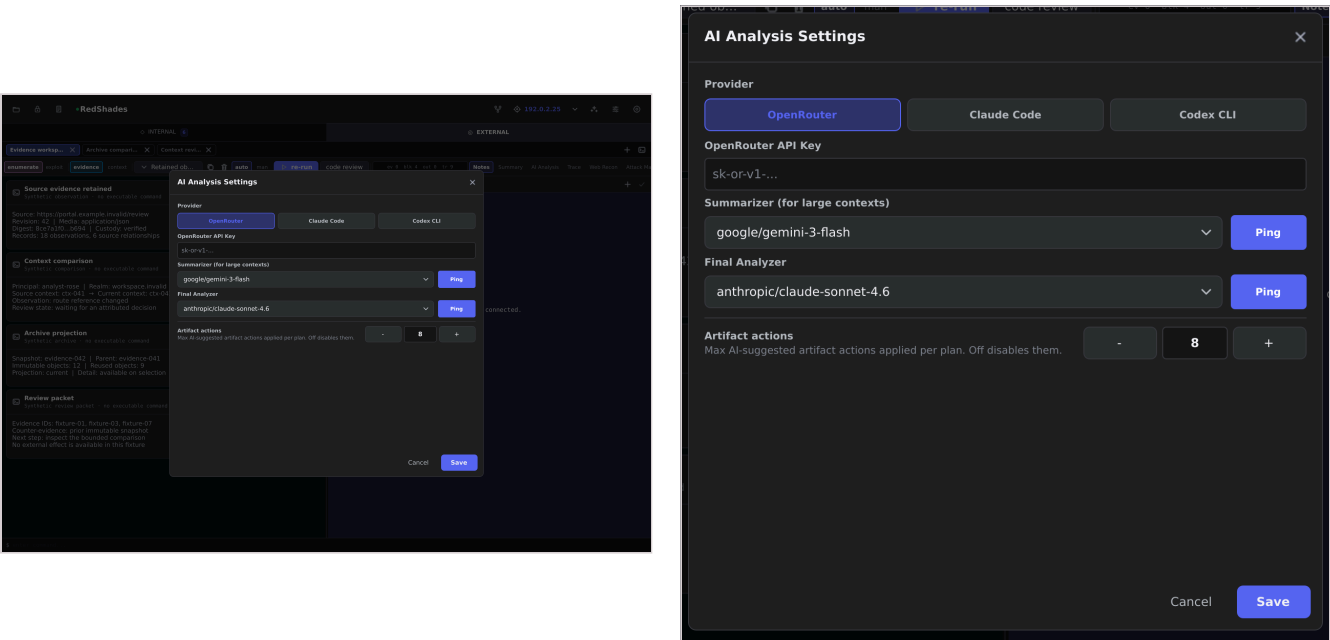
Reference 19

hot swappable stage workers

WHAT DOES CLEANUP NEED TO KNOW?

Context and resource budgets stay visible

Custom context management and token accounting track the cost of model work. Resource supervision retains ownership of worker processes, bounded output and cleanup, supporting long sessions without losing sight of their compute footprint.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Overview and detail crop from identical source pixels. Presentation evidence only.

Token and resource diagnostics make unusual consumption inspectable. Named owners and retained cleanup handles let the platform distinguish active work from stale work.

Implemented supervision boundaries

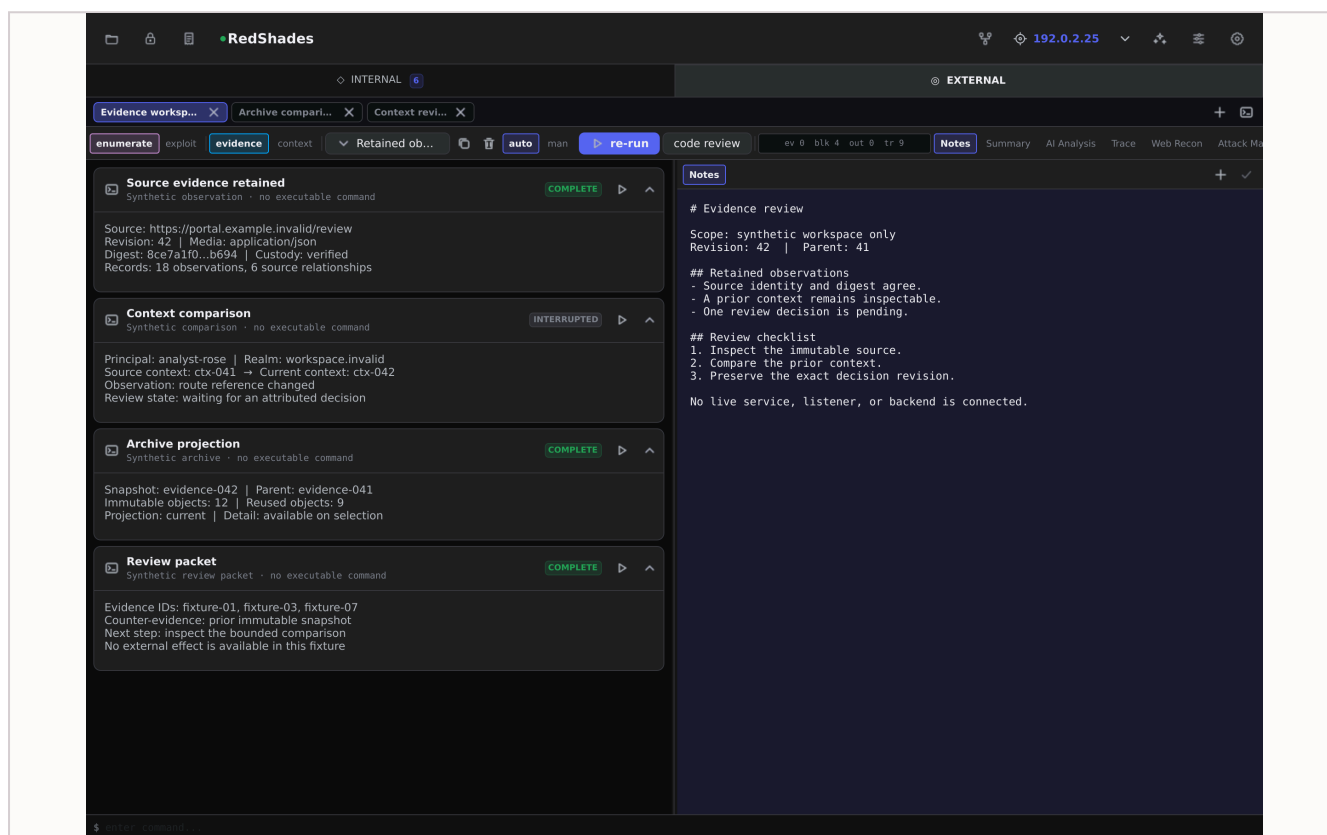
Reference 20

service process custody

HOW THE WORKSPACE FITS TOGETHER

Qt, watchdog and tools in one workspace

The native Qt operator places dense graphs and investigation panels inside a managed desktop workspace. The watchdog hosts retained service and tool surfaces, connecting Gitea, BloodHound, build workflows, RDP and development harnesses with the operator environment.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

Lightweight QML chrome surrounds native widgets. Embedded tool windows and custom analysis views can sit alongside each other, preserving tool fidelity while keeping the investigation visible.

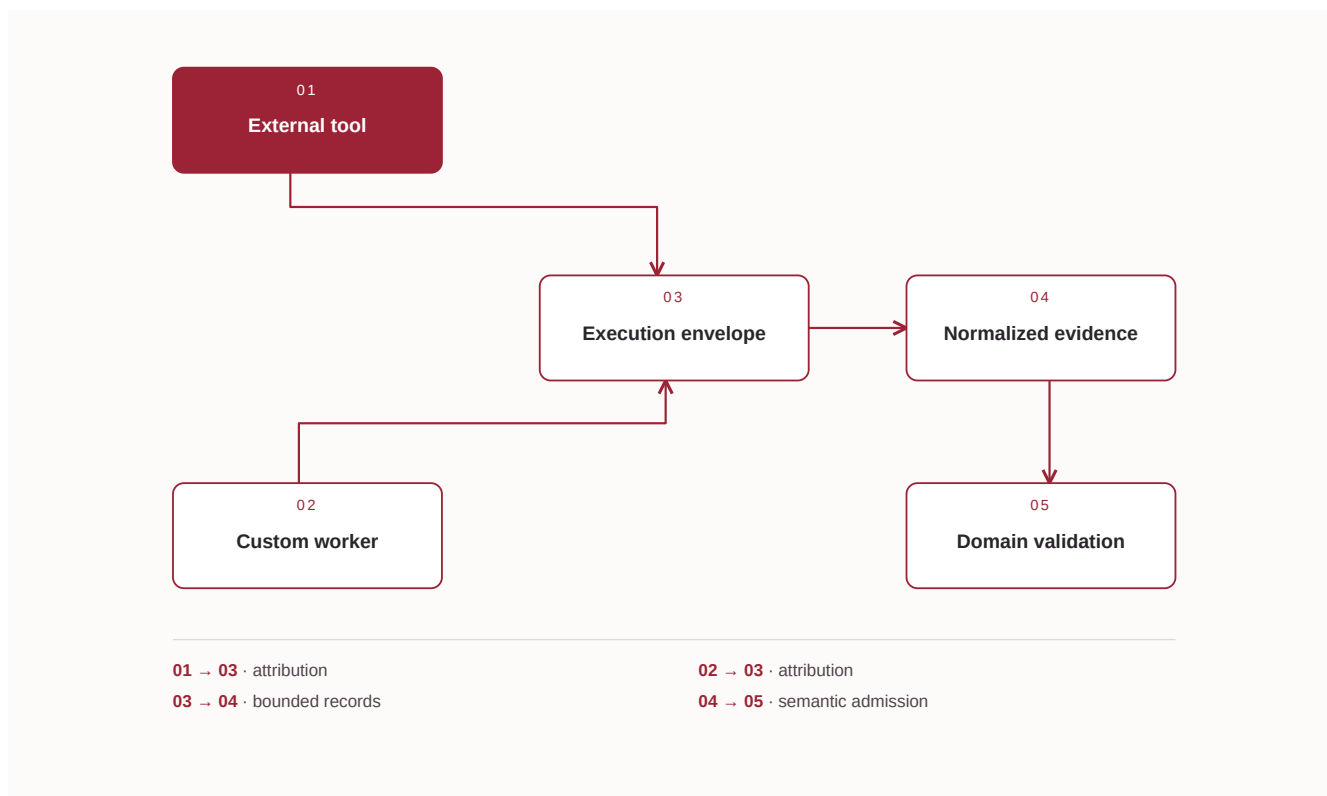
Implemented contract

Reference 21
subsystem glossary

WHAT MAKES A TOOL CATALOG MORE THAN A LIST?

Third-party tools, isolated execution

Tool descriptors connect network, web, directory, browser and source-inspection families to the platform. Admitted third-party proof-of-concept work can use isolated container environments, with outputs returned as attributed evidence.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Integration combines execution policy, process ownership and evidence normalization. Tool availability, candidate review and permission remain distinct inputs to the workflow.

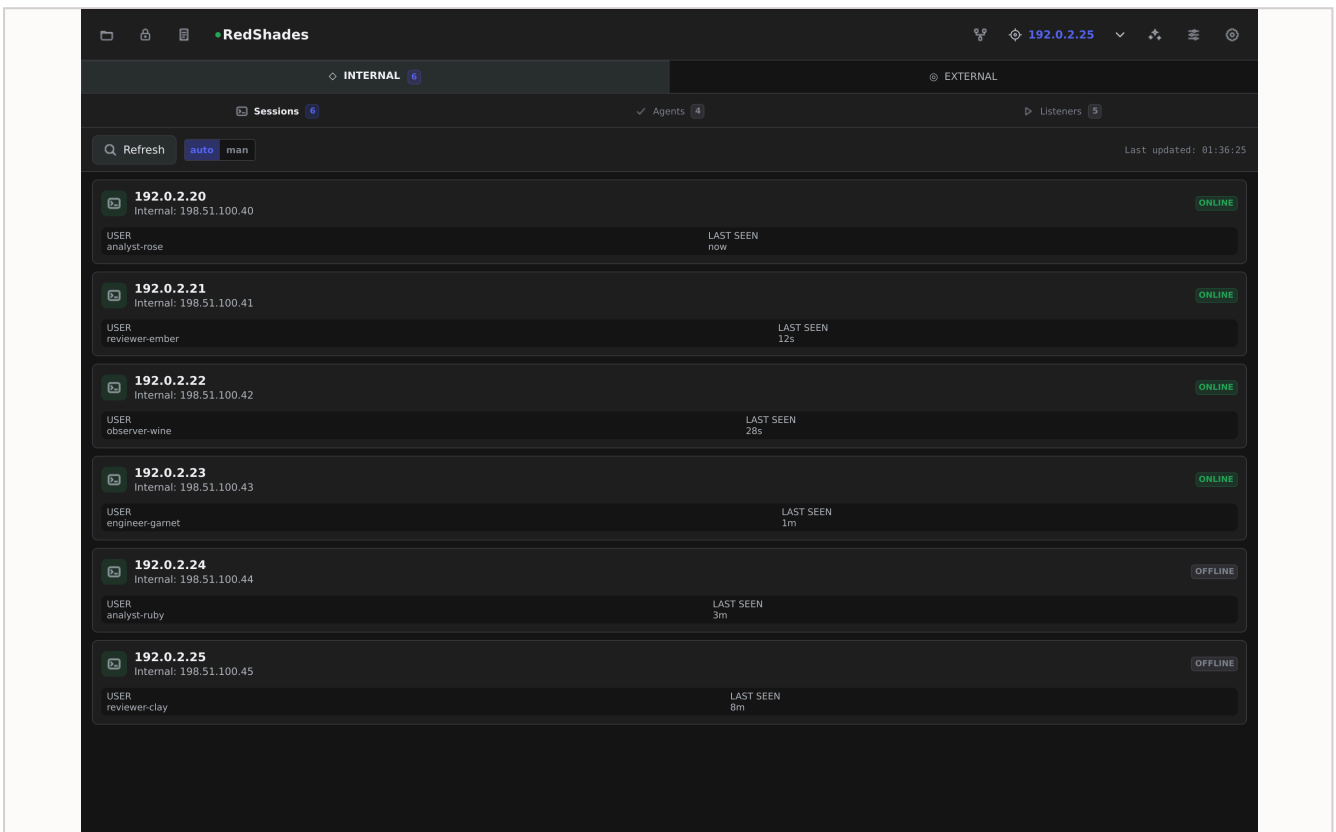
Implemented contract

Reference 22
specialist plugin registry

WHAT BINDS A BINARY TO ITS USE?

VM-based transformation and artifact engineering

The artifact and implant toolchain includes transformation and obfuscation work alongside versioned builds. VM-oriented transformation is part of the project's engineering breadth, with artifact identity and provenance carried into the wider workspace.



Actual Qt capture · 5 September 2026 · fictional, disconnected fixture. Presentation evidence only.

The publication describes these systems at component level. Build records identify the material and its role; native artifact surfaces keep the resulting inventory accessible to the operator.

Implemented contract

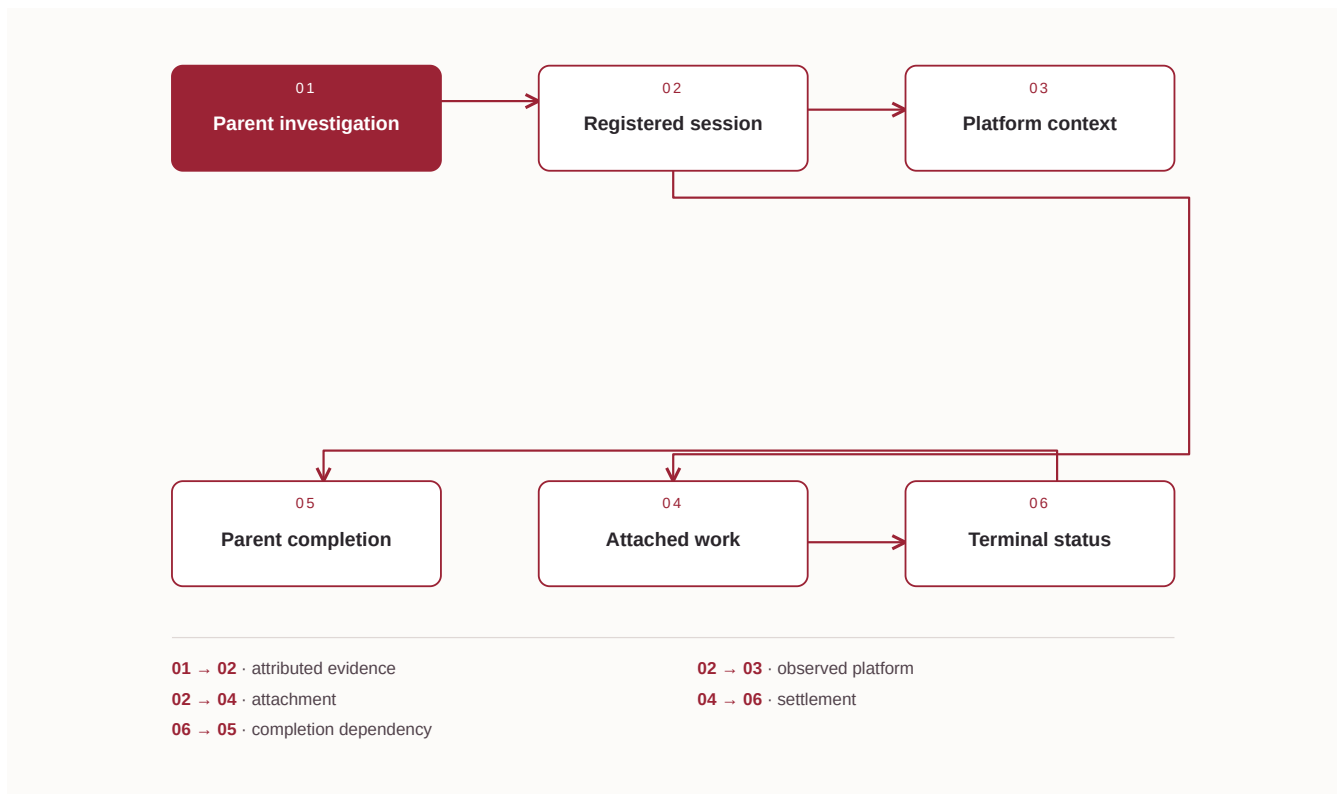
Reference 23

deployment and process ownership

HOW IS POST-FOOTHOLD WORK REPRESENTED?

Session work extends the investigation

C2 stage families add platform-aware continuation once a session is registered. Context, artifacts and attached work remain connected to the parent engagement, extending the same orchestration model beyond initial observation.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Common and Windows/Linux declarations preserve their own prerequisites. The automatic stage atlas and the C2 declaration inventory describe separate parts of the runtime graph.

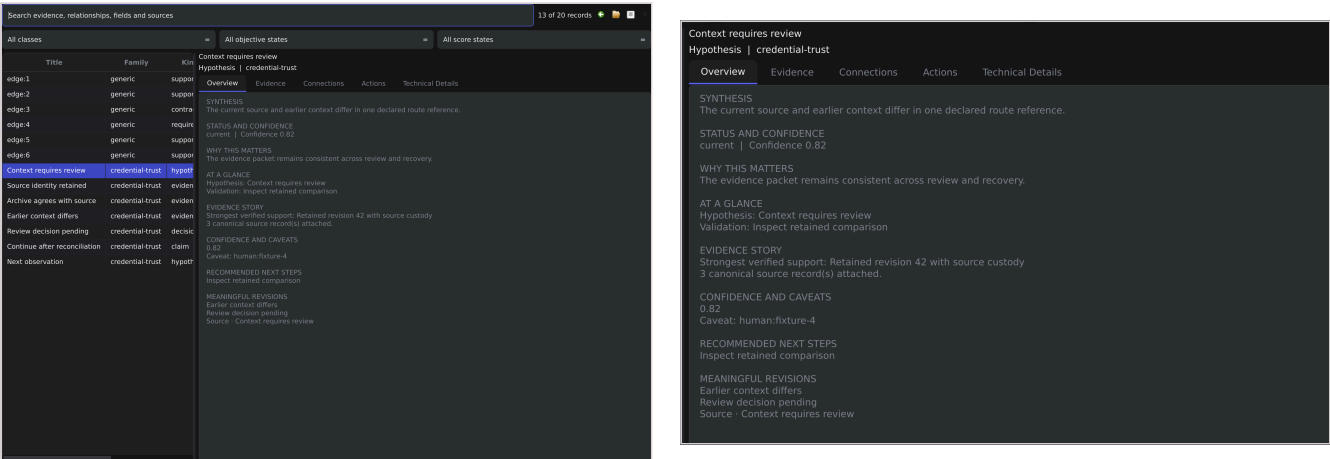
Implemented contract

Reference 24
 authoritative engagement architecture

EMBEDDED SPECIALIST TOOLS AND CUSTOM ANALYSIS

Reverse engineering inside the workflow

Binary and source analysis connect to the investigation workbench and embedded specialist tools. Custom reverse-engineering workflows can sit beside native evidence views and retained external tool surfaces.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Overview and detail crop from identical source pixels. Presentation evidence only.

The integrated workspace brings analysis, artifact records and engineering context together. Tool-specific interfaces remain available while their results feed the shared investigation record.

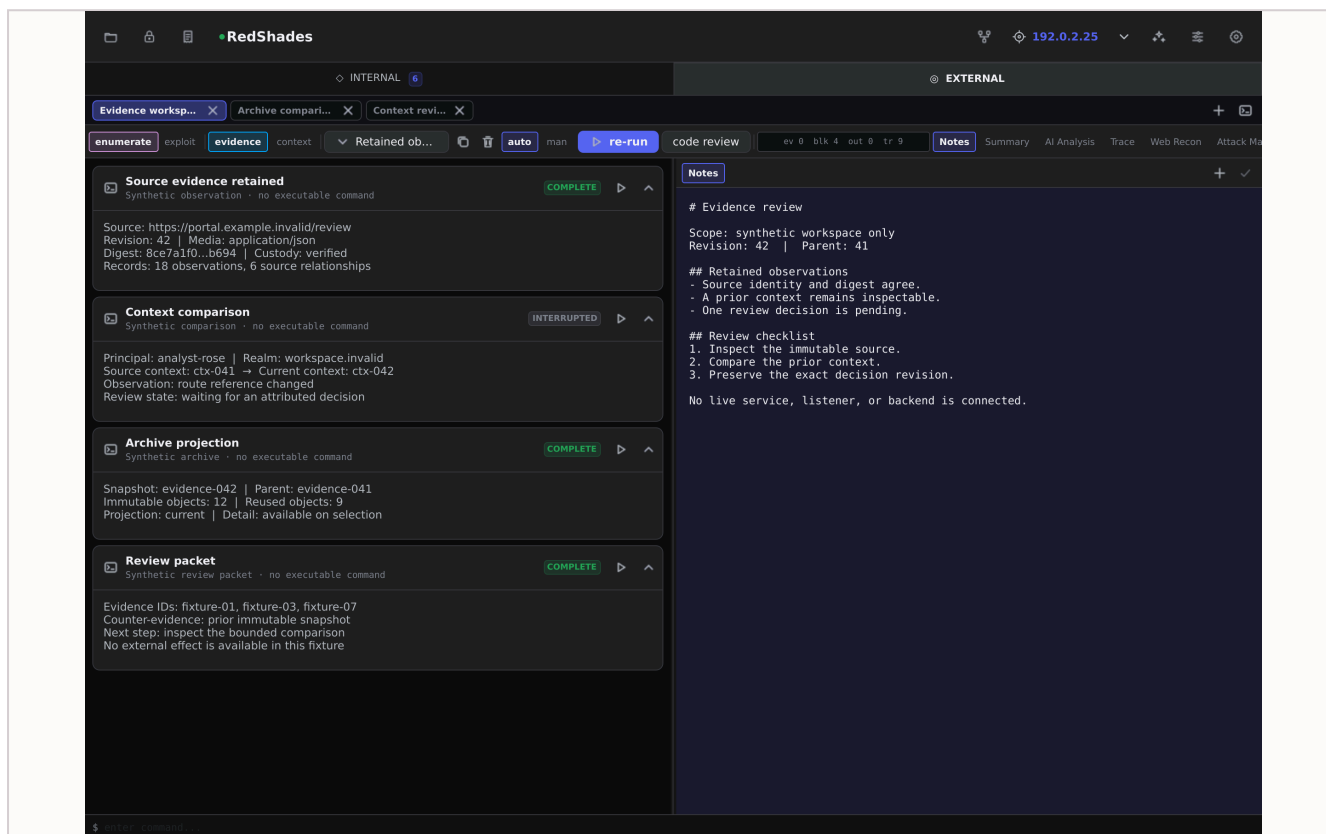
Implemented contract

Reference 25
execution graph lifecycle

WHAT SHOULD AN OPERATOR BE ABLE TO EXPLAIN?

A working desktop for the whole investigation

Network Graph, WebRecon, Investigation Graph, session records and model controls share a native Qt operator. Embedded tools extend the workspace rather than forcing every workflow into a generic dashboard.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

The retained screenshots show real Qt widgets with fictional disconnected data. Their arrangement demonstrates how overview, selected evidence and specialist work can remain visible together.

Sanitized native UI evidence

Reference 26

subsystem glossary

NETWORK GRAPH

Read the investigation spatially

The Network Graph presents observed hosts and relationships spatially. Its value is orientation: which context is selected and how that context relates to the wider investigation.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

Native Qt capture with fictional, disconnected demonstration data. The web edition opens the original image at full resolution.

Sanitized native UI evidence

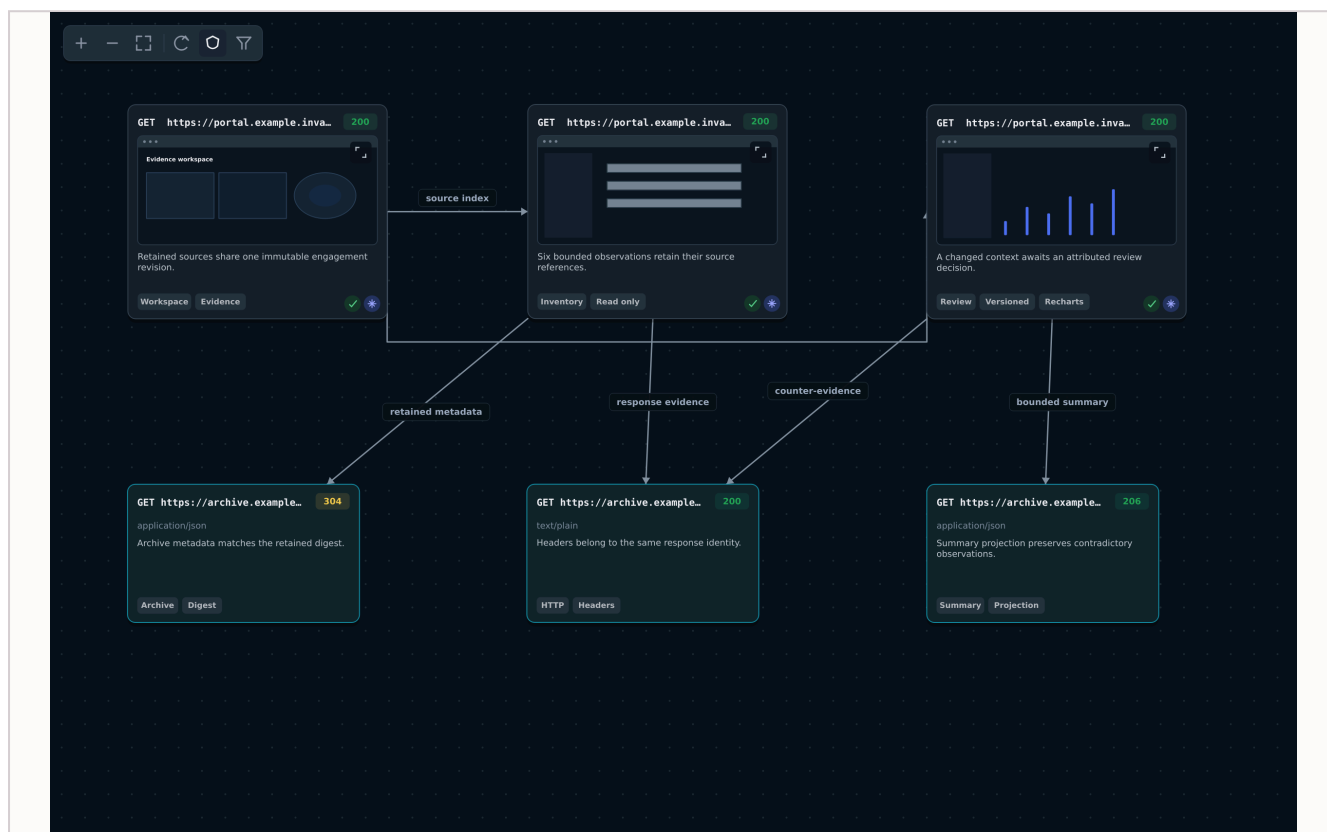
Reference 27

subsystem glossary

WEBRECON OVERVIEW

Keep the source near the decision

WebRecon brings web observations, source detail and endpoint review into a native workspace. A useful arrangement lets the operator retain overview while examining the evidence behind one selected item.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

Native Qt capture with fictional, disconnected demonstration data. The web edition opens the original image at full resolution.

Sanitized native UI evidence

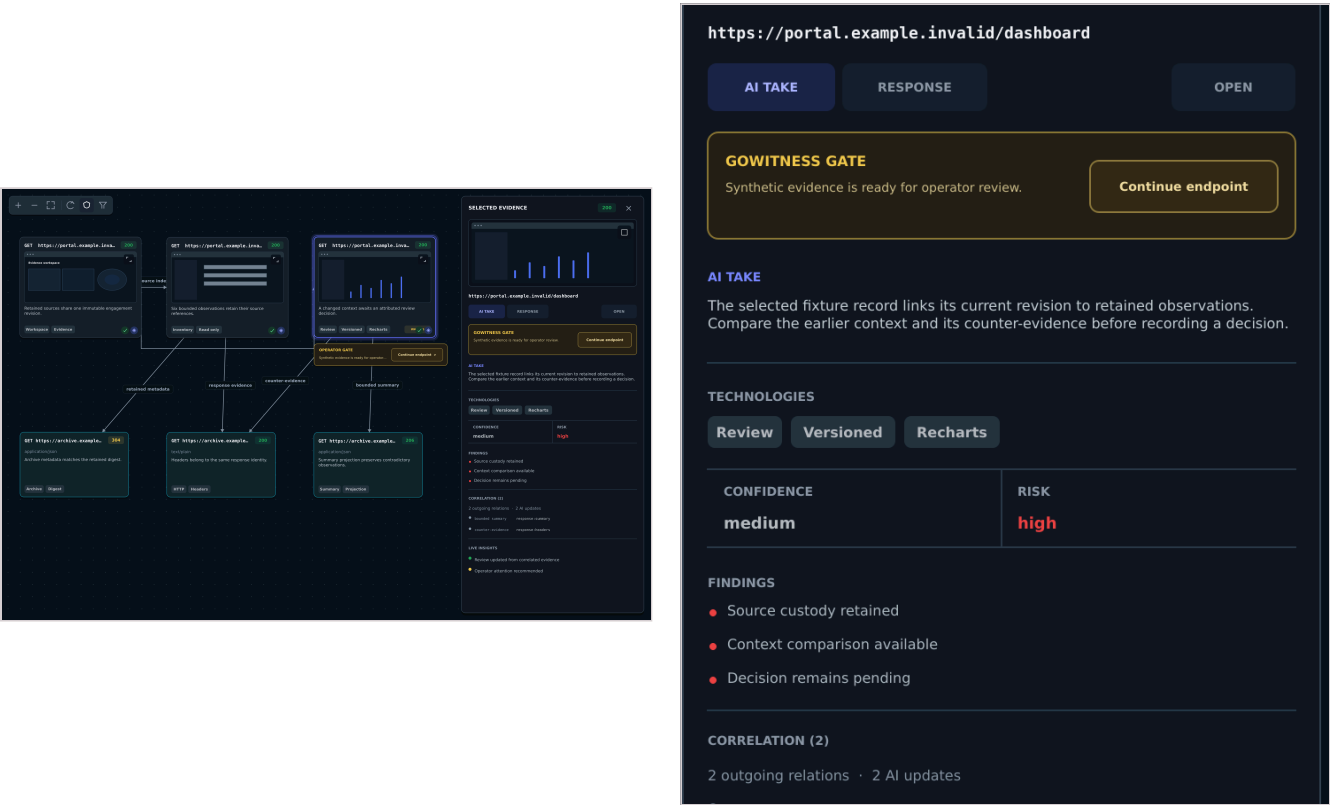
Reference 28

canonical specialist contracts

WEBRECON REVIEW AND GATE STATE

A pending decision has a scope

A gate should explain the affected endpoint or branch and retain the evidence needed for a decision. The rest of the run should not appear globally blocked unless it was explicitly paused.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Overview and detail crop from identical source pixels. Presentation evidence only.

Native Qt capture with fictional, disconnected demonstration data. The web edition opens the original image at full resolution.

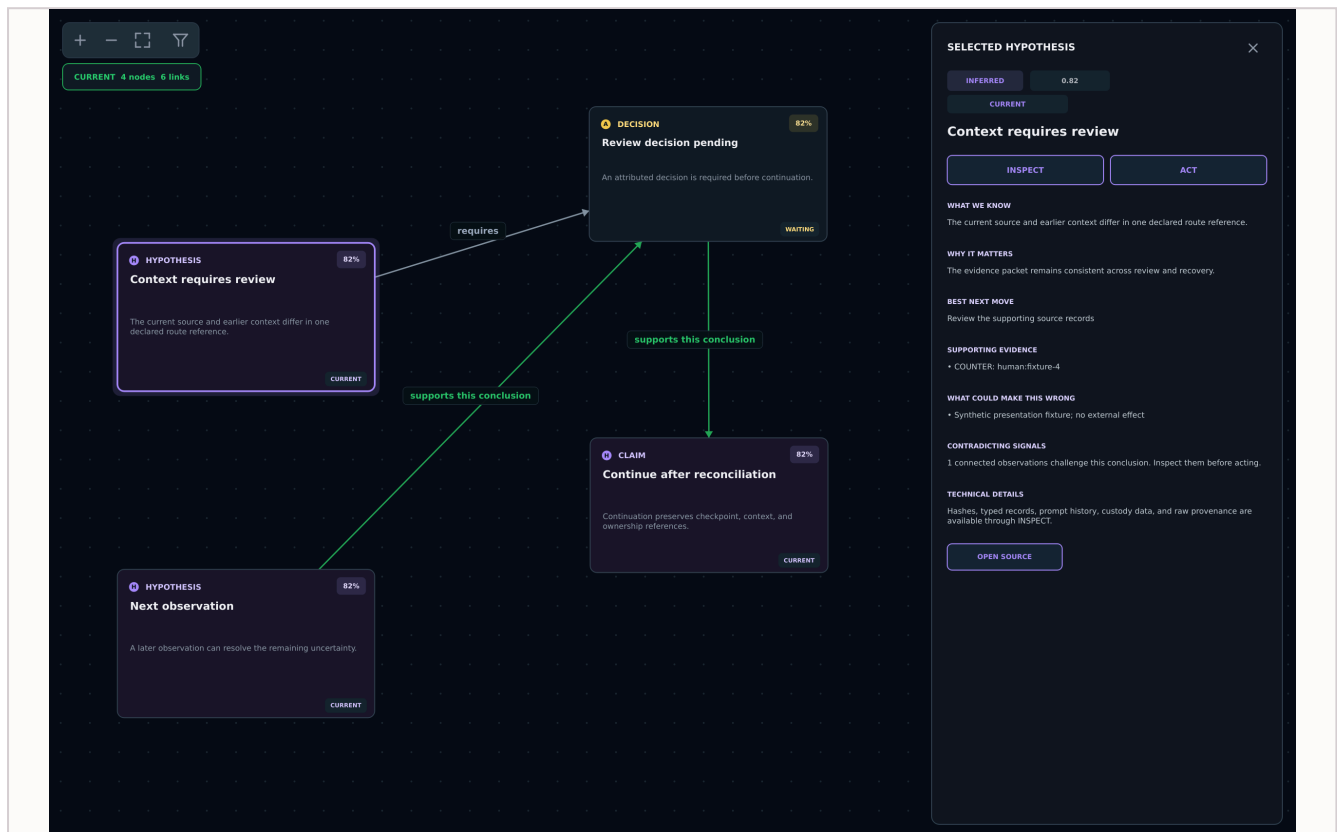
Sanitized native UI evidence

Reference 29
execution graph lifecycle

INVESTIGATION GRAPH

Reasoning needs inspectable sources

Investigation Graph organizes relationships among observations and interpretations. Its companion detail views preserve source references so the operator can inspect the basis of a hypothesis.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

Native Qt capture with fictional, disconnected demonstration data. The web edition opens the original image at full resolution.

Sanitized native UI evidence

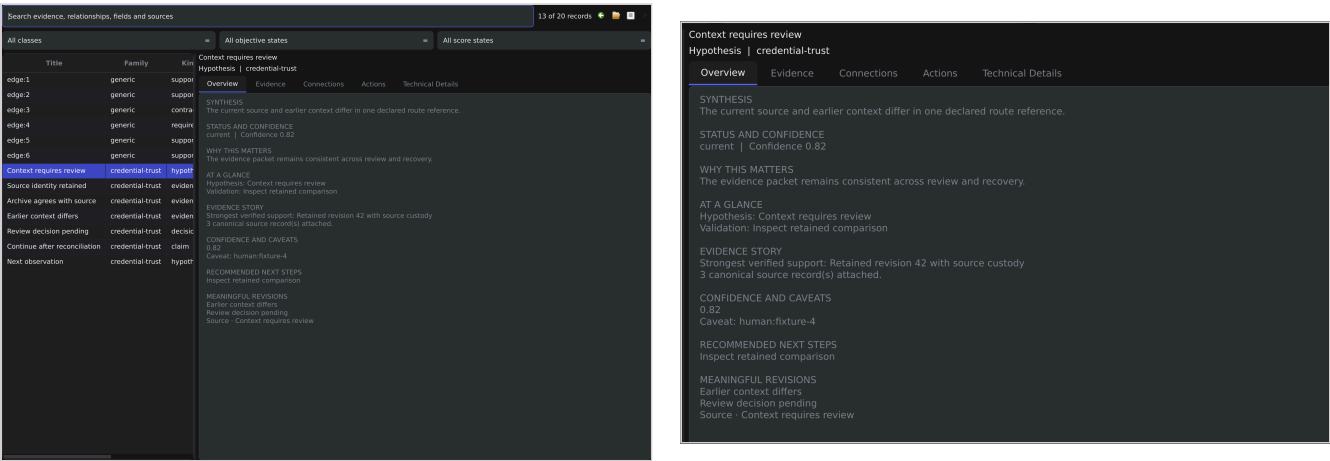
Reference 30

investigation attention

INVESTIGATION WORKBENCH

The detail view carries the difficult questions

The Workbench provides a more deliberate place to inspect evidence, context and available continuation. Dense records benefit from selection and detail rather than forcing every field into the graph itself.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Overview and detail crop from identical source pixels. Presentation evidence only.

Native Qt capture with fictional, disconnected demonstration data. The web edition opens the original image at full resolution.

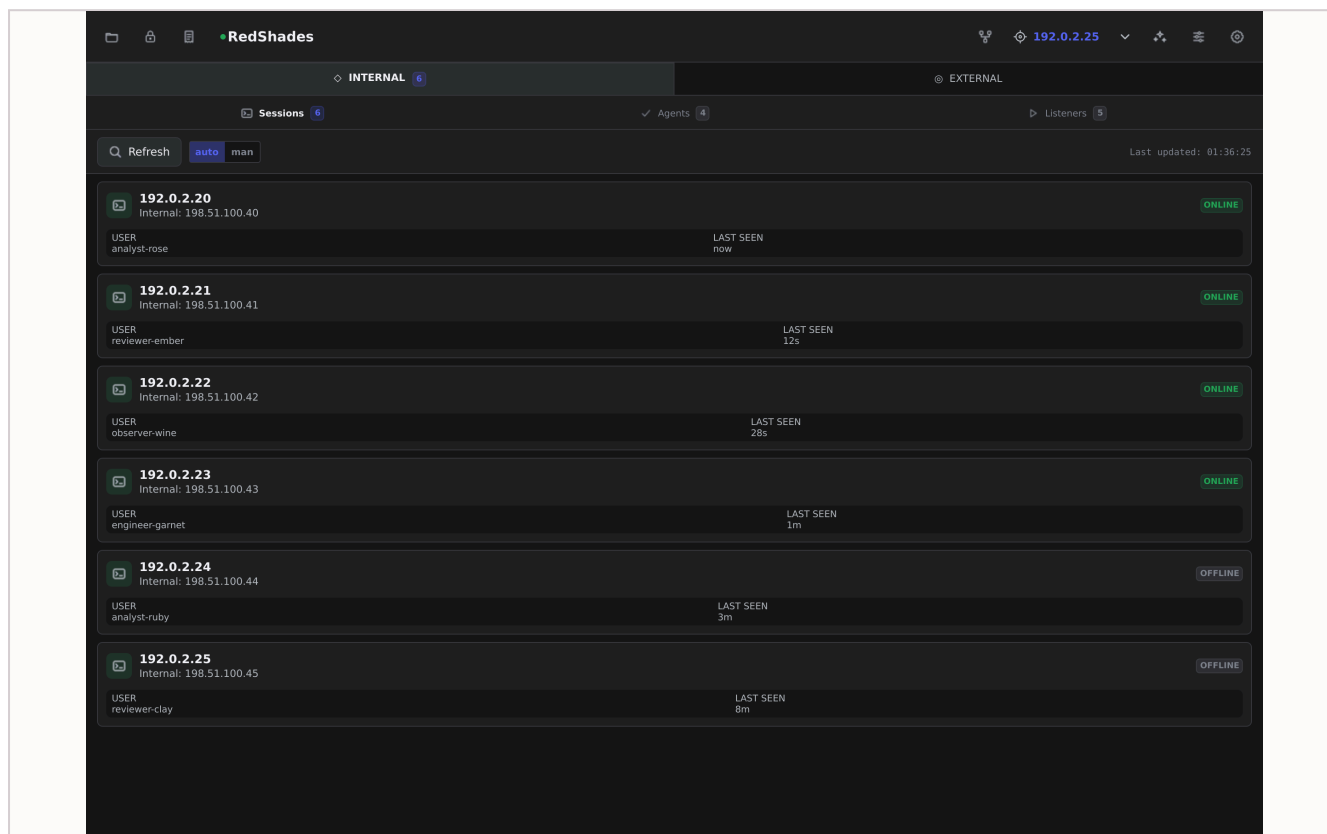
Sanitized native UI evidence

Reference 31
canonical specialist contracts

SESSION AND ARTIFACT SURFACES

Sessions and artifacts at a glance

Session and agent views bring stored records, versioned builds and observed availability into the operator workspace. Selected detail connects the inventory with its context and current state.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

Native Qt capture with fictional, disconnected demonstration data. The web edition opens the original image at full resolution.

Sanitized native UI evidence

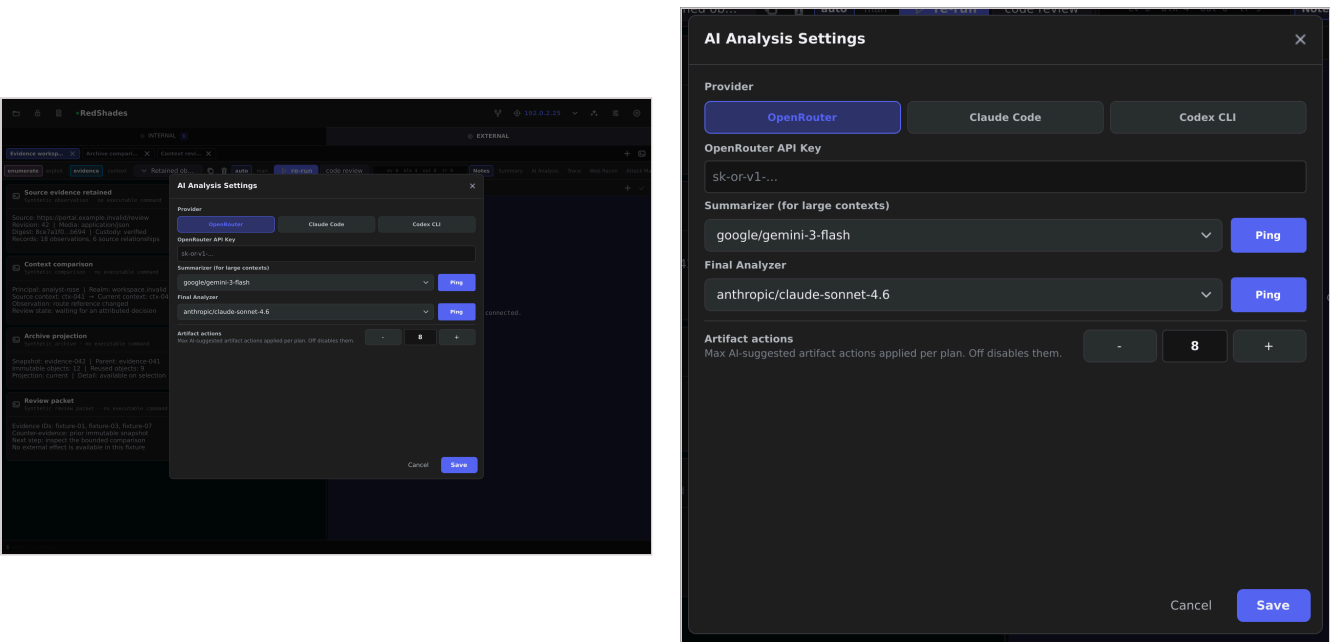
Reference 32

subsystem glossary

MODEL AND RUNTIME CONTROLS

Model and runtime controls in reach

Operator settings expose model and runtime choices in the same workspace as the investigation. Configuration surfaces connect human intent with backend admission, attribution and accounting.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Overview and detail crop from identical source pixels. Presentation evidence only.

Native Qt capture with fictional, disconnected demonstration data. The web edition opens the original image at full resolution.

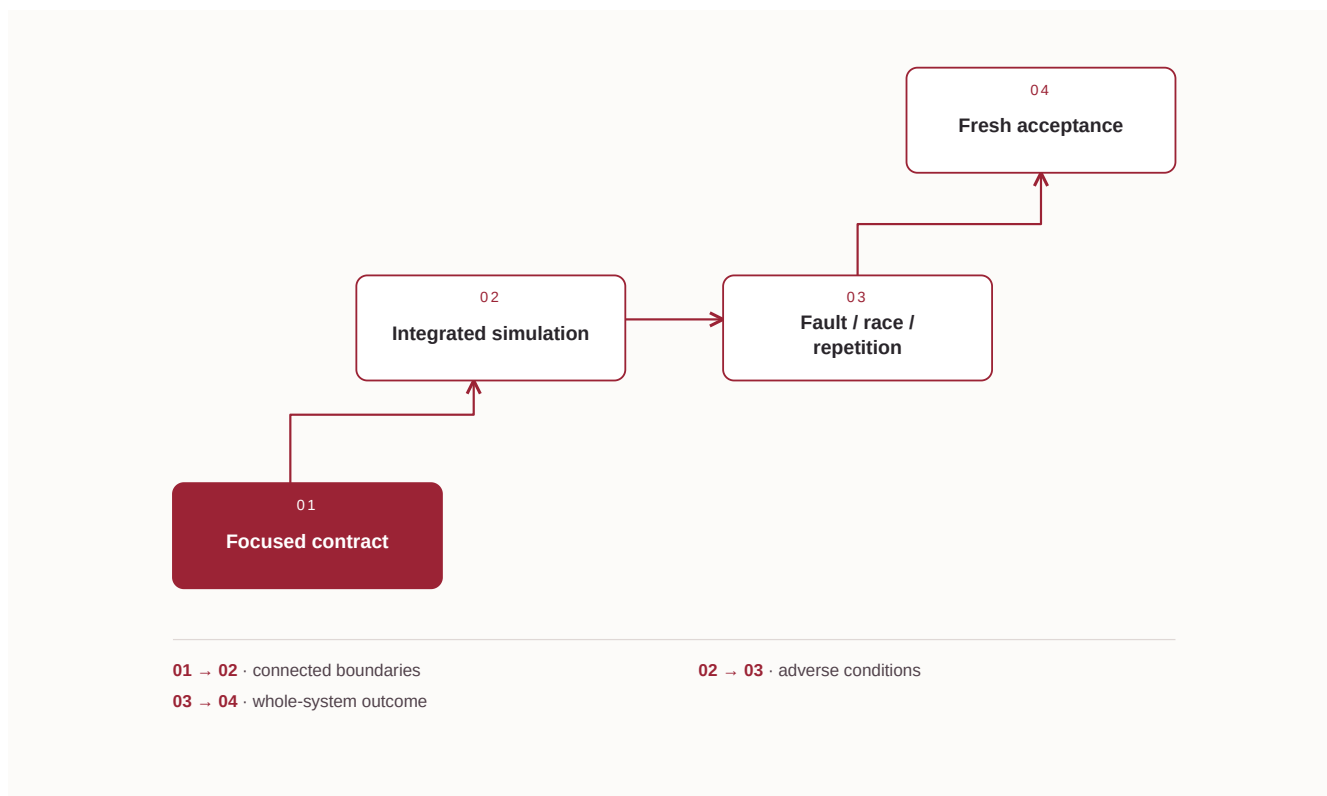
Sanitized native UI evidence

Reference 33
authoritative engagement architecture

HOW SHOULD CONFIDENCE ACCUMULATE?

Verification at every scale

Focused contracts, integrated simulations, race checks, repetition, fuzzing and fault injection exercise different dimensions of the platform. Mega and Ultra organize these into explicit engineering acceptance work.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Revision-bound records connect each check to its command and output. Validation status is documented separately from the product's architectural capabilities.

Implemented contract

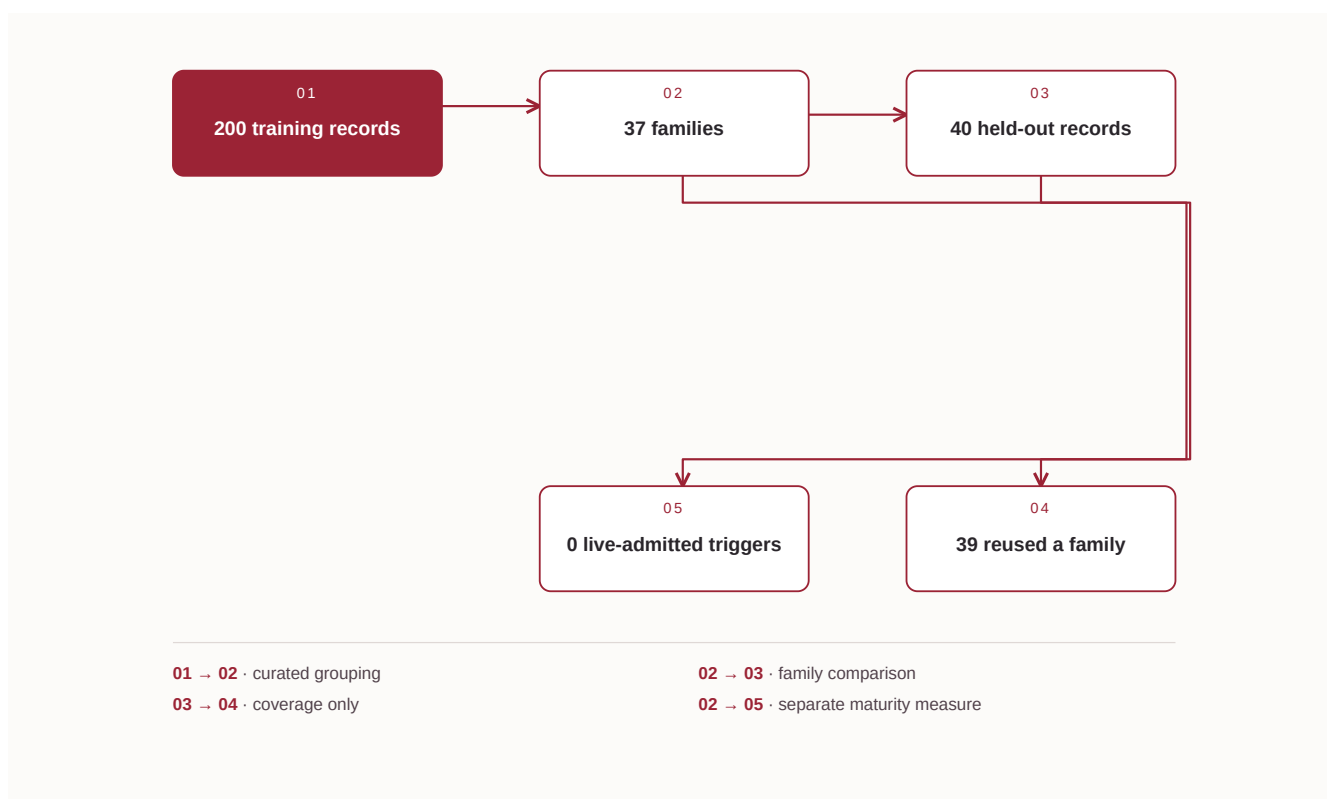
Reference 34

ultra ci requirements

WHAT DID THE DISCLOSURE STUDY MEASURE?

Disclosures become a reusable methodology vocabulary

The Trigger Methodology Graph study organized 200 public disclosures into 37 recurring families. In a 40-disclosure identity-disjoint holdout, 39 belonged to a family represented in that vocabulary.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

This is family coverage in a curated study, not a count of executed discoveries. It motivates evidence-bound hypotheses and controlled follow-up through the diagnostic graph.

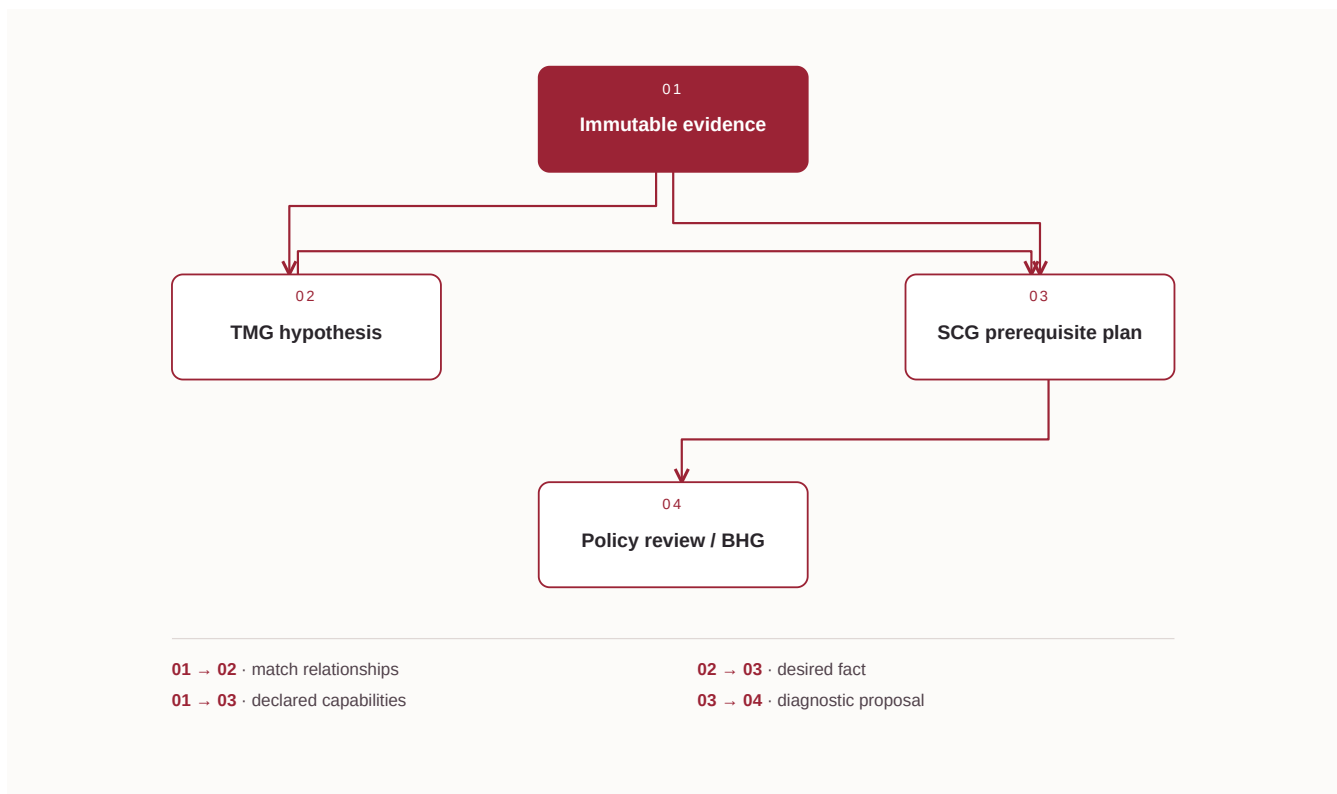
Retained study

Reference 35
tmg disclosure generalization study

HOW DO TMG AND SCG CONTRIBUTE?

A hypothesis should reveal its missing evidence

Trigger Methodology Graph (TMG) produces evidence-bound diagnostic hypotheses. Shadow Capability Graph (SCG) reasons about missing prerequisites from an immutable snapshot. Bounty Hunting Gates (BHG) retain policy review for proposed hypotheses.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Their useful role is to make uncertainty and the next required evidence explicit. A diagnostic graph does not become scheduling or execution authority simply because its suggestion is plausible.

Implemented contract

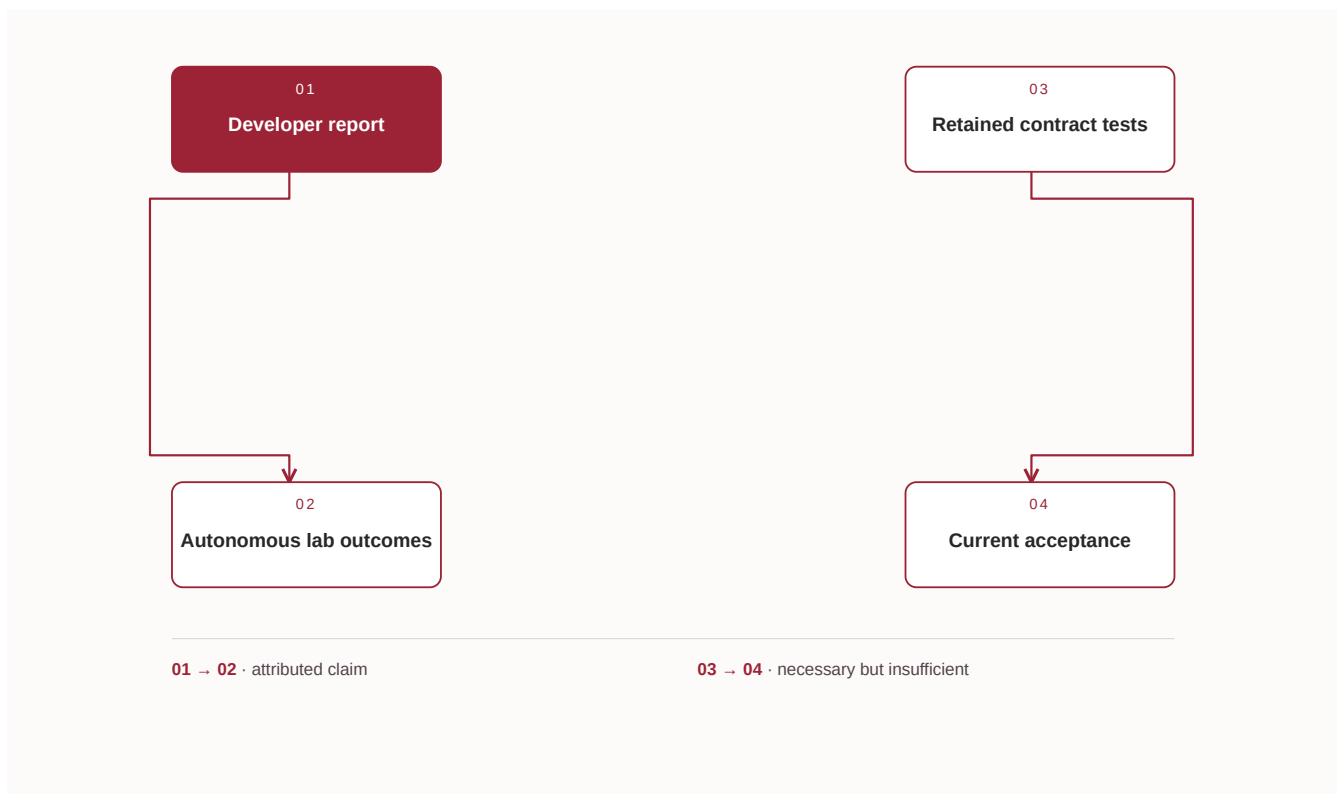
Reference 36

trigger methodology graph + shadow capability graph

WHAT HAS BEEN REPORTED?

Autonomous outcomes across Windows and web labs

The developer reports unattended completion of P.O.O, a hard Windows CTF, an insane Windows HTB lab and roughly two easy Linux/web HTB labs. These outcomes reflect the project's ambition to carry substantial work across an entire engagement.



Source-derived conceptual diagram. Arrows describe the labeled relationship.

Results are owner-reported autonomous outcomes. Independent complete log bundles were not established by this editorial review; the reported runs are not presented as a benchmark for every unseen environment.

Developer-reported autonomous outcomes

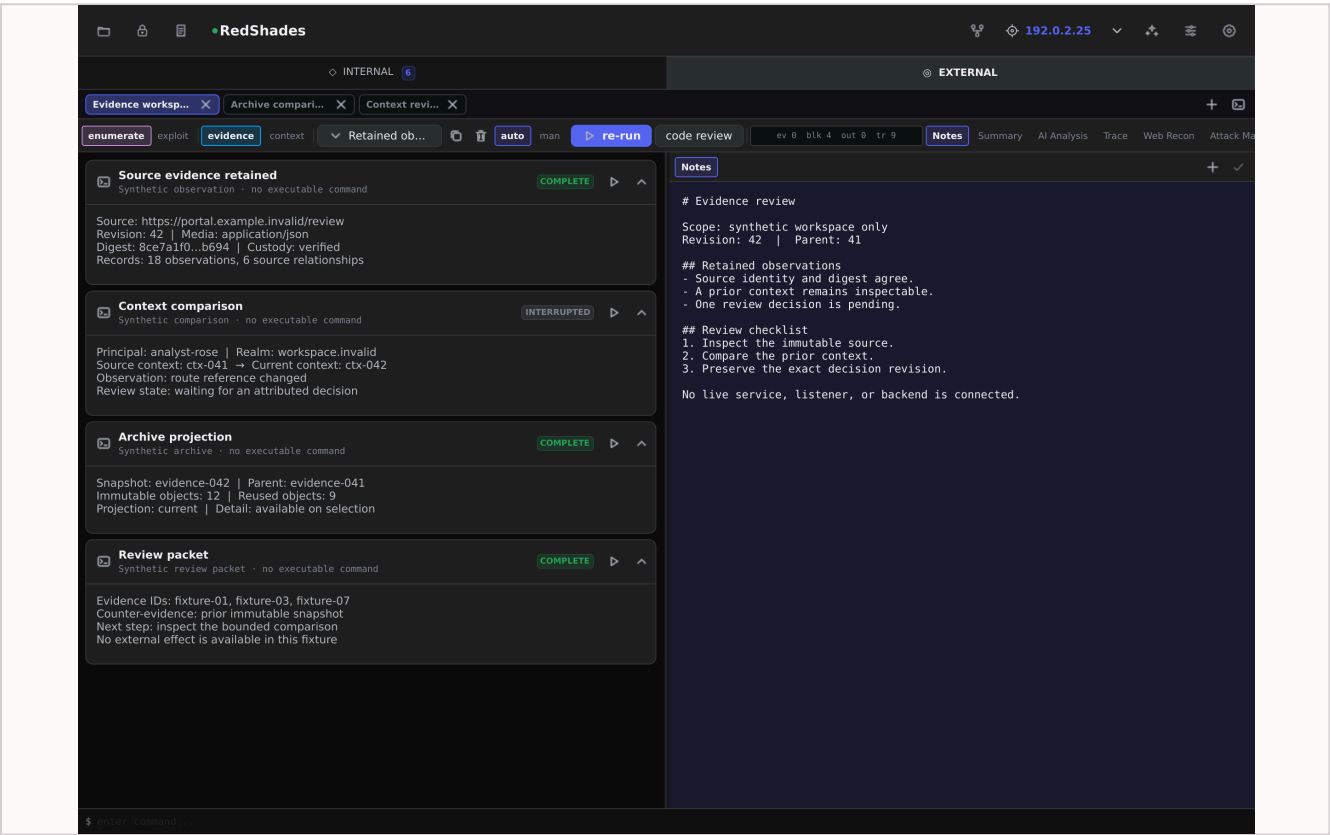
Reference 37

design history

AN INVITATION TO EXAMINE THE BOUNDARIES

Explore the system in a private demonstration

RedShades brings adaptive orchestration, custom capability engineering, runtime repair and a native tool workspace into one product architecture. The investment proposition is the combination: a platform built to make focused model intelligence useful across substantial technical work.



Actual Qt capture - 5 September 2026 - fictional, disconnected fixture. Presentation evidence only.

A private technical demonstration can examine the operator, follow a retained capability through review and explore the engineering behind persistent sessions and hot replacement. Independent technology, presented for technical investors and product leaders.

Technical portfolio

Reference 38
editorial thesis

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

implant deployment Validated Core`implant_deployment`

Artifact and session work · Admitted artifact and authorized context → Artifact/session records

ACL operator review (post-summary) Validated Core`acl_operator_action`

Directory analysis · Directory evidence and operator review → Review record

AD compositional capability planning Validated Core`ad_capability_planning`

Directory analysis · Directory evidence → Directory analysis records

AD effective capability discovery Validated Core`ad_capability_discovery`

Directory analysis · Directory evidence → Directory analysis records

BloodHound collection Validated Core`bloodhound_collection`

Directory analysis · Directory context → Graph analysis material

Cross-realm Kerberos service-ticket custody Validated Core`ad_cross_realm_kerberos`

Directory analysis · Directory evidence → Directory analysis records

masscan discovery Validated Core`discovery_masscan`

Discovery · Scope and observation context → Network observations

nmap service and OS scan Validated Core`discovery_nmap`

Discovery · Scope and observation context → Network observations

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

passive OSINT (WHOIS / DNS / ASN) Validated Core`osint_passive`**Discovery** • Scope and public observations → Context evidence**query discovered ports** Validated Core`discovery_ports`**Discovery** • Scope and observation context → Network observations**BloodHound ACL attack path analysis** Validated Core`post_acl_analysis`**Extended investigation** • Session, artifact or investigation context → Attributed investigation records**chisel SOCKS5 reverse pivot** Validated Core`post_pivot_chisel`**Extended investigation** • Session, artifact or investigation context → Attributed investigation records**DCSync domain hash dump** Validated Core`post_dcsync`**Extended investigation** • Session, artifact or investigation context → Attributed investigation records**Disabled legacy advisory handoff** Validated Core`post_nopac`**Extended investigation** • Session, artifact or investigation context → Attributed investigation records**DPAPI artifact analysis (operator action)** Validated Core`post_dpapi`**Extended investigation** • Session, artifact or investigation context → Attributed investigation records**Final assessment report** Validated Core`post_pwndoc_export`**Extended investigation** • Session, artifact or investigation context → Attributed investigation records

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

Linux privilege escalation enumeration Validated Core`post_linux_privesc`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

pass-the-hash sweep Validated Core`post_pth`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

pass-the-ticket (golden/silver ticket) Validated Core`post_ptt`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

secretsdump credential harvest Validated Core`post_secretsdump`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

SMB binary credential extraction Validated Core`post_smb_binary_analysis`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

Windows privilege escalation enumeration Validated Core`post_windows_privesc`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

AD coercion (PrinterBug/PetitPotam/DFSCoerce) Validated Core`identity_coerce`

Identity analysis • Identity and service context → Identity evidence

certipy AD CS Validated Core`identity_certipy`

Identity analysis • Identity and service context → Identity evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

certipy AD CS exploitation (ESC1/ESC4/ESC8) Validated Core

identity_certipy_exploit

Identity analysis • Identity and service context → Identity evidence

enum4linux-ng Validated Core

identity_enum4linux

Identity analysis • Identity and service context → Identity evidence

GetNPUsers Validated Core

identity_getnpusers

Identity analysis • Identity and service context → Identity evidence

GetUserSPNs Validated Core

identity_getuserspns

Identity analysis • Identity and service context → Identity evidence

hash cracking Validated Core

identity_hash_cracking

Identity analysis • Identity and service context → Identity evidence

kerbrute Validated Core

identity_kerbrute

Identity analysis • Identity and service context → Identity evidence

LAPS password dump Validated Core

identity_laps

Identity analysis • Identity and service context → Identity evidence

ldapdomaindump Validated Core

identity_ldapdomaindump

Identity analysis • Identity and service context → Identity evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

lookupsid Validated Core`identity_lookupsid`

Identity analysis • Identity and service context → Identity evidence

NTLM relay (ntlmrelayx) Validated Core`identity_ntlm_relay`

Identity analysis • Identity and service context → Identity evidence

nxc null/guest enumeration Validated Core`identity_nxc_guest`

Identity analysis • Identity and service context → Identity evidence

Responder NTLM hash capture Validated Core`identity_ntlm_capture`

Identity analysis • Identity and service context → Identity evidence

targeted credential spray Validated Core`identity_credential_spray`

Identity analysis • Identity and service context → Identity evidence

username permutation generation Validated Core`identity_username_anarchy`

Identity analysis • Identity and service context → Identity evidence

credential reuse sweep Validated Core`credential_reuse_sweep`

Investigation • Scoped evidence → Attributed records

Execution context: network path and domain time Validated Core`environment_adaptation`

Platform adaptation • Environment context → Adaptation records

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

conditional mssqlclient Validated Core`remote_mssqlclient`

Remote services • Service and authorized session context → Protocol evidence

conditional smbclient Validated Core`remote_smbclient`

Remote services • Service and authorized session context → Protocol evidence

Kerberos ticket-backed WinRM validation Validated Core`remote_kerberos_winrm`

Remote services • Service and authorized session context → Protocol evidence

MSSQL linked server ADIDNS poisoning Validated Core`mssql_linked_server_poison`

Remote services • Database context → Database evidence

SMB file harvest Validated Core`remote_smb_harvest`

Remote services • Service and authorized session context → Protocol evidence

smbmap share enumeration Validated Core`remote_smbmap`

Remote services • Service and authorized session context → Protocol evidence

deterministic operator report Validated Core`summary`

Reporting • Retained investigation evidence → Report projection

Disabled legacy service-effect handoff Validated Core`service_redis`

Service analysis • Service observations → Service evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

Disabled legacy service-effect handoff Validated Core`service_mysql`

Service analysis • Service observations → Service evidence

Disabled legacy service-effect handoff Validated Core`service_ipmi`

Service analysis • Service observations → Service evidence

DNS zone transfer drill Validated Core`service_dns_axfr`

Service analysis • Service observations → Service evidence

dnsx Validated Core`service_dnsx`

Service analysis • Service observations → Service evidence

FTP anonymous Validated Core`service_ftp_anon`

Service analysis • Service observations → Service evidence

NFS enumeration Validated Core`service_nfs`

Service analysis • Service observations → Service evidence

service CVE candidate sweep Validated Core`service_searchsploit_sweep`

Service analysis • Service observations → Service evidence

service CVE workbench Validated Core`service_cve_workbench`

Service analysis • Service observations → Service evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

SMTP user enumeration Validated Core`service_smtp_user_enum`

Service analysis · Service observations → Service evidence

SNMP enumeration Validated Core`service_snmp_enum`

Service analysis · Service observations → Service evidence

ssh-audit Validated Core`service_ssh_audit`

Service analysis · Service observations → Service evidence

subfinder Validated Core`service_subfinder`

Service analysis · Service observations → Service evidence

credential to shell attempts Validated Core`shell_attempts`

Session work · Authorized execution context → Session-related records

xfreerdp RDP session Validated Core`rdp_session`

Session work · Authorized session context → Session records

CMS scanning Validated Core`web_cms_scan`

Web investigation · Web observations and scoped review → Web evidence

CVE PoC workbench review Validated Core`web_cve_poc_workbench`

Web investigation · Web observations and scoped review → Web evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

Disabled legacy web-effect handoff Validated Core`web_cms_exploit`

Web investigation · Web observations and scoped review → Web evidence

Disabled legacy web-effect handoff Validated Core`web_sql_i_post_exploit`

Web investigation · Web observations and scoped review → Web evidence

Disabled legacy web-effect handoff Validated Core`web_rce_confirm`

Web investigation · Web observations and scoped review → Web evidence

GoWitness endpoint review Validated Core`web_screenshot`

Web investigation · Web observations and scoped review → Web evidence

HTTP verb tampering Validated Core`web_verb_tampering`

Web investigation · Web observations and scoped review → Web evidence

IDOR / parameter enumeration Validated Core`web_idor_probe`

Web investigation · Web observations and scoped review → Web evidence

LFI exploitation Validated Core`web_lfi_exploit`

Web investigation · Web observations and scoped review → Web evidence

nuclei parallel Validated Core`web_nuclei`

Web investigation · Web observations and scoped review → Web evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

OS command injection probe Validated Core`web_cmdi_probe`

Web investigation · Web observations and scoped review → Web evidence

post-auth CVE PoC workbench review Validated Core`web_post_auth_poc_workbench`

Web investigation · Web observations and scoped review → Web evidence

post-auth exploit probes Validated Core`web_post_auth_exploit_probes`

Web investigation · Web observations and scoped review → Web evidence

post-auth surface harvest Validated Core`web_post_auth_harvest`

Web investigation · Web observations and scoped review → Web evidence

SQLi probe Validated Core`web_sql_i_probe`

Web investigation · Web observations and scoped review → Web evidence

SSTI probe Validated Core`web_ssti_probe`

Web investigation · Web observations and scoped review → Web evidence

target-specific password generation Validated Core`web_cewl`

Web investigation · Web observations and scoped review → Web evidence

web content analysis Validated Core`web_content_analysis`

Web investigation · Web observations and scoped review → Web evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Exact source labels and keys. Input and output classes are descriptive summaries; separate C2 families extend session work.

web file upload attack Validated Core`web_file_upload_attack`

Web investigation · Web observations and scoped review → Web evidence

web fingerprint + wafw00f Validated Core`web_whatweb_waf`

Web investigation · Web observations and scoped review → Web evidence

web fuzzing (vhost+dir+ext) Validated Core`web_fuzz`

Web investigation · Web observations and scoped review → Web evidence

web login brute force Validated Core`web_login_bruteforce`

Web investigation · Web observations and scoped review → Web evidence

web RCE candidate correlation Validated Core`web_rce_candidate_correlation`

Web investigation · Web observations and scoped review → Web evidence

web RCE deployment preparation Validated Core`web_rce_deployment_prepare`

Web investigation · Web observations and scoped review → Web evidence

XSS probe Validated Core`web_xss_probe`

Web investigation · Web observations and scoped review → Web evidence

XXE probe Validated Core`web_xxe_probe`

Web investigation · Web observations and scoped review → Web evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

PLANNED INTEGRATIONS · PACKAGING PENDING

Clusters share capabilities and tools.

48 descriptors: 37 potential third-party packages, one external Apple toolchain and ten native or custom components. Domain totals overlap through reuse.

Cluster → capability → shared tool architecture

Domain cluster	Analysis capabilities	Primary and alternative components
Android / mobile 8 stages · 10 referenced entries	artifact intake · provenance custody · configuration review · static analysis · topology modeling · evidence correlation · emulation simulation · reporting	Androguard · Apktool · Android bundletool · Cosign · JADX · Mobile Security Framework · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades evidence-backed reporting
Apple platforms 8 stages · 8 referenced entries	artifact intake · provenance custody · binary decomposition · configuration review · static analysis · topology modeling · emulation simulation · reporting	ipsw · LLVM binary utilities · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades privacy/data-flow model · RedShades evidence-backed reporting · Apple codesign/security toolchain
Blockchain 8 stages · 8 referenced entries	artifact intake · provenance custody · static analysis · evidence correlation · emulation simulation · topology modeling · reporting	Cosign · Echidna · Foundry · RedShades artifact intake · RedShades typed evidence graph · RedShades evidence-backed reporting · Slither · Solidity compiler
Bluetooth / BLE / NFC 7 stages · 8 referenced entries	artifact intake · passive capture parsing · configuration review · evidence correlation · topology modeling · reporting	BlueZ btmon · libnfc utilities · RedShades artifact intake · RedShades typed evidence graph · RedShades protocol observation normalizer · RedShades evidence-backed reporting · Wireshark/TShark · Zeek
Wi-Fi / RF 8 stages · 9 referenced entries	artifact intake · topology modeling · passive capture parsing · static analysis · evidence correlation · policy gating · reporting	GNU Radio · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades protocol observation normalizer · RedShades evidence-backed reporting · Suricata · Wireshark/TShark · Zeek
LLM / AI systems 8 stages · 10 referenced entries	artifact intake · topology modeling · configuration review · emulation simulation · evidence correlation · supply chain analysis · reporting	garak · Inspect AI · OSV-Scanner · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades privacy/data-flow model · RedShades evidence-backed reporting · Syft · Trivy

95 specialist stages: 78 Advanced and 17 Ultimate. Licensing and validation apply to each integration. The 48 entries include 47 stage-referenced components and one foundational router.

Under testing / Under development

22 open-source integration candidates · 15 license-review candidates · 5 built-in · 5 custom planned · 1 external

PLANNED INTEGRATIONS · PACKAGING PENDING

Clusters share capabilities and tools.

48 descriptors: 37 potential third-party packages, one external Apple toolchain and ten native or custom components.
Domain totals overlap through reuse.

Cluster → capability → shared tool architecture

Domain cluster	Analysis capabilities	Primary and alternative components
Satellite / space 7 stages · 10 referenced entries	artifact intake · passive capture parsing · configuration review · topology modeling · emulation simulation · reporting	GNU Radio · gr-satellites · Orekit · RedShades artifact intake · RedShades typed evidence graph · RedShades protocol observation normalizer · RedShades evidence-backed reporting · RedShades OT/space safety model · Wireshark/TShark · Zeek
Quantum readiness 6 stages · 6 referenced entries	cryptographic inventory · evidence correlation · configuration review · topology modeling · emulation simulation · reporting	Cirq · Qiskit SDK · RedShades artifact intake · RedShades cryptographic evidence normalizer · RedShades typed evidence graph · RedShades evidence-backed reporting
Hardware / firmware 9 stages · 11 referenced entries	artifact intake · binary decomposition · supply chain analysis · static analysis · configuration review · evidence correlation · emulation simulation · reporting	Binwalk · Cosign · Ghidra · ipsw · LLVM binary utilities · QEMU · RedShades artifact intake · RedShades typed evidence graph · RedShades evidence-backed reporting · Syft · Trivy
ICS / SCADA 9 stages · 11 referenced entries	artifact intake · passive capture parsing · topology modeling · safety modeling · configuration review · static analysis · evidence correlation · emulation simulation · policy gating	OpenSCAP · QEMU · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades protocol observation normalizer · RedShades evidence-backed reporting · RedShades OT/space safety model · Suricata · Wireshark/TShark · Zeek
Defensive patching 8 stages · 9 referenced entries	supply chain analysis · evidence correlation · static analysis · reporting · policy gating	Cosign · Grype · OSV-Scanner · RedShades typed evidence graph · RedShades defensive change planner · RedShades policy and effect gate · RedShades evidence-backed reporting · Syft · Trivy
Digital forensics 9 stages · 12 referenced entries	provenance custody · artifact intake · forensic analysis · evidence correlation · reporting	Cosign · Plaso · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades evidence-backed reporting · The Sleuth Kit · Velociraptor · Volatility 3 · Wireshark/TShark · YARA · Zeek

95 specialist stages: 78 Advanced and 17 Ultimate. Licensing and validation apply to each integration. The 48 entries include 47 stage-referenced components and one foundational router.

Under testing / Under development

22 open-source integration candidates · 15 license-review candidates · 5 built-in · 5 custom planned · 1 external

UNDER TESTING

Specialist application domains

Android / mobile

8 stages

Under testing

Entry evidence

Application and device evidence

Conditional branches

Package, permission and storage branches

Capabilities

Mobile configuration, application metadata and transport review

Shared tool architecture

10 referenced entries · Androguard, Apktool, Android bundletool, Cosign, JADX, Mobile Security Framework. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Apple platforms

8 stages

Under testing

Entry evidence

iOS, iPhone and macOS evidence

Conditional branches

Application, entitlement and platform branches

Capabilities

Signing, entitlements, privacy settings and platform configuration

Shared tool architecture

8 referenced entries · ipsw, LLVM binary utilities, Apple codesign/security toolchain. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Blockchain

8 stages

Under testing

Entry evidence

Contract and transaction evidence

Conditional branches

Contract, dependency and governance branches

Capabilities

Contract metadata, access controls and transaction provenance

Shared tool architecture

8 referenced entries · Cosign, Echidna, Foundry, Slither, Solidity compiler. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Core: 88 stages · Advanced: 78 additional stages · Ultimate: 17 additional stages.

 Under testing

UNDER TESTING

Connected and intelligent systems

Proximity / BLE / NFC

7 stages

Under testing

Entry evidence

Captured proximity observations

Conditional branches

Protocol and device-class branches

Capabilities

Advertisement, pairing and NFC record analysis

Shared tool architecture

8 referenced entries · BlueZ btmon, libnfc utilities, Wireshark/TShark, Zeek. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Wi-Fi / RF

8 stages

Under testing

Entry evidence

Wireless configuration and recorded traffic

Conditional branches

Network, radio and authentication branches

Capabilities

Wireless posture, spectrum records and authentication configuration

Shared tool architecture

9 referenced entries · GNU Radio, Suricata, Wireshark/TShark, Zeek. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

LLM / AI systems

8 stages

Under testing

Entry evidence

Model, application and evaluation evidence

Conditional branches

Data, model and application boundary branches

Capabilities

Evaluation artifacts, data handling and permission boundaries

Shared tool architecture

10 referenced entries · garak, Inspect AI, OSV-Scanner, Syft, Trivy. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Core: 88 stages · Advanced: 78 additional stages · Ultimate: 17 additional stages.

Under testing

UNDER TESTING

Infrastructure at the edge

Satellite / space

7 stages

Under testing

Entry evidence

Ground-system and telemetry artifacts

Conditional branches

Ground, link and telemetry branches

Capabilities

Telemetry formats, asset configuration and trust boundaries

Shared tool architecture

10 referenced entries · GNU Radio, gr-satellites, Orekit, Wireshark/TShark, Zeek. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Quantum readiness

6 stages

Under testing

Entry evidence

Cryptographic asset inventory

Conditional branches

Algorithm, dependency and migration branches

Capabilities

Cryptographic discovery and migration readiness analysis

Shared tool architecture

6 referenced entries · Cirq, Qiskit SDK. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Core: 88 stages · Advanced: 78 additional stages · Ultimate: 17 additional stages.

Under testing

UNDER TESTING

Physical and industrial systems

Hardware / firmware

9 stages

Under testing

- Entry evidence
- Firmware images and component inventory
- Conditional branches
- Image, component and boot-policy branches
- Capabilities
- Firmware metadata, component provenance and secure-boot posture
- Shared tool architecture
- 11 referenced entries · Binwalk, Cosign, Ghidra, ipsw, LLVM binary utilities, QEMU, Syft, Trivy. Native evidence and reporting components complete the cluster.
- Gate & convergence
- Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

ICS / SCADA

9 stages

Under testing

- Entry evidence
- Industrial asset and configuration evidence
- Conditional branches
- Asset, protocol and zone branches
- Capabilities
- Configuration review, protocol records and segmentation posture
- Shared tool architecture
- 11 referenced entries · OpenSCAP, QEMU, Suricata, Wireshark/TShark, Zeek. Native evidence and reporting components complete the cluster.
- Gate & convergence
- Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Core: 88 stages · Advanced: 78 additional stages · Ultimate: 17 additional stages.

Under testing

UNDER DEVELOPMENT

PurpleShades and digital forensics

PurpleShades • Defensive patching

8 stages • **Under development**

Asset and finding intake opens parallel exposure, dependency and remediation analysis. A change plan brings the branches together with rollback criteria, canary observations and approval requirements.

Advisories → assessment branches → change proposal → rollback / canary criteria → approval → review packet

The workflow assembles a reviewed proposal and defines the evidence required to validate a future change.

Digital forensics

9 stages • **Under development**

Case intake establishes evidence identity and custody. Artifact, timeline and correlation branches preserve source attribution and converge into a reviewable case record.

Case intake → evidence preservation → analysis branches → correlation → review → case report

Integrity checks precede analysis. Incomplete or conflicting observations remain visible in the final evidence account.

Core: 88 stages • Advanced: 78 additional stages • Ultimate: 17 additional stages.

Under development

ANNUAL PRICING PER SEAT

Three bundles. One connected practice.

Core

Starting at

\$7,000

/ year per seat

88 Core stages

Validated Core

Advanced

Starting at

\$14,000

/ year per seat

Core + 78 specialist stages

Under testing

Ultimate

Starting at

\$20,000

/ year per seat

Advanced + 17 defensive and forensic stages

Under development

183 stages in total. Separate C2 session declarations are counted independently.

Core: 88 stages · Advanced: 78 additional stages · Ultimate: 17 additional stages.

Annual pricing per seat